KiPI power-integrity report

Audio_Visualiser_DSP.kicad_pcb · 2026-10-18 21:21 · KiPI 0.9.0 · solved in 290.8 s

Mesh: max edge 0.3 mm, via grading 0.2, copper resistivity 1.724e-08 Ω·m at 20 °C (TCR 0.00393/°C). Via plating, first matching rule (drill = hole in the laminate, plated inward): micro any drill: 25 um, if filled: non-conductive epoxy; buried any drill: 25 um, if filled: non-conductive epoxy; blind any drill: 25 um, if filled: non-conductive epoxy; blind_buried any drill: 25 um, if filled: non-conductive epoxy; through any drill: 25 um, if filled: non-conductive epoxy; pth any drill: 25 um, if filled: non-conductive epoxy; any any drill: 25 um, if filled: non-conductive epoxy.

Stackup, top to bottom: top silkscreen 15 µm, top solder mask 10 µm, F.Cu 35 µm, 3313 RC57% 99.4 µm εr 4.1, In1.Cu 15.2 µm, FR4 core 100 µm εr 4.6, In2.Cu 15.2 µm, 2116 RC54% x3 356.8 µm εr 4.16, In3.Cu 15.2 µm, FR4 core 300 µm εr 4.6, In4.Cu 15.2 µm, 2116 RC54% x3 356.8 µm εr 4.16, In5.Cu 15.2 µm, FR4 core 100 µm εr 4.6, In6.Cu 15.2 µm, 3313 RC57% 99.4 µm εr 4.1, B.Cu 35 µm, bottom solder mask 10 µm, bottom silkscreen 15 µm; 1.574 mm in all.

3D Results View: Audio_Visualiser_DSP_kipi_report_2026-10-18_211614_3d.html

Loads

Load	Group	I (A)	Nominal (V)	V effective (V)	Drop / boost (mV)	%	Worst pin (V)	Worst pin (mV)	Worst %	Tolerance	Result
J2101	+12V (+2 nets)	1.944	12	11.812072	187.928 drop	1.566	11.811973	188.027 drop	1.567	±5 %	PASS
J2102	+12V (+2 nets)	1.944	12	11.801674	198.326 drop	1.653	11.801615	198.385 drop	1.653	±5 %	PASS
U701 dsPIC VDD	+3.3V (+3 nets)	0.0346	3.3	3.298678	1.322 drop	0.040	3.298597	1.403 drop	0.043	±3 %	PASS
U701 dsPIC AVDD	+3.3V (+3 nets)	0.0074	3.3	3.296653	3.347 drop	0.101	3.296653	3.347 drop	0.101	±3 %	PASS
X601 oscillator	+3.3V (+3 nets)	0.007	3.3	3.228543	71.457 drop	2.165	3.228543	71.457 drop	2.165	±3 %	PASS
U1301 MCP6N11	+3.3V (+3 nets)	0.0011	3.3	3.299180	0.820 drop	0.025	3.299179	0.821 drop	0.025	±3 %	PASS
U1302 MCP6N11	+3.3V (+3 nets)	0.0011	3.3	3.299094	0.906 drop	0.027	3.299094	0.906 drop	0.027	±3 %	PASS
U1401 MCP6N11	+3.3V (+3 nets)	0.0011	3.3	3.299464	0.536 drop	0.016	3.299463	0.537 drop	0.016	±3 %	PASS
U1402 MCP6N11	+3.3V (+3 nets)	0.0011	3.3	3.299286	0.714 drop	0.022	3.299285	0.715 drop	0.022	±3 %	PASS
U1502 MCP6024	+3.3V (+3 nets)	0.0054	3.3	3.299042	0.958 drop	0.029	3.299042	0.958 drop	0.029	±3 %	PASS
U1601 MCP6024	+3.3V (+3 nets)	0.0054	3.3	3.298867	1.133 drop	0.034	3.298867	1.133 drop	0.034	±3 %	PASS
U1701 MCP6024	+3.3V (+3 nets)	0.0054	3.3	3.298974	1.026 drop	0.031	3.298974	1.026 drop	0.031	±3 %	PASS
U1501 TS3A24159	+3.3V (+3 nets)	5e-05	3.3	3.299167	0.833 drop	0.025	3.299162	0.838 drop	0.025	±3 %	PASS
U1503 MCP4451	+3.3V (+3 nets)	0.0006	3.3	3.299080	0.920 drop	0.028	3.299042	0.958 drop	0.029	±3 %	PASS
U1702 REF35	+3.3V (+3 nets)	3e-06	3.3	3.298966	1.034 drop	0.031	3.298966	1.034 drop	0.031	±3 %	PASS
U801 PCA9536	+3.3V (+3 nets)	0.00029	3.3	3.298586	1.414 drop	0.043	3.298586	1.414 drop	0.043	±3 %	PASS
U803 TCA9801	+3.3V (+3 nets)	0.00275	3.3	3.298621	1.379 drop	0.042	3.298621	1.379 drop	0.042	±3 %	PASS
U1807 TCA9801	+3.3V (+3 nets)	0.00275	3.3	3.298509	1.491 drop	0.045	3.298509	1.491 drop	0.045	±3 %	PASS
U1809 TCA9801	+3.3V (+3 nets)	0.00275	3.3	3.298422	1.578 drop	0.048	3.298422	1.578 drop	0.048	±3 %	PASS
U1803 74VHC541	+3.3V (+3 nets)	0.001	3.3	3.298492	1.508 drop	0.046	3.298490	1.510 drop	0.046	±3 %	PASS
U1804 74VHC541	+3.3V (+3 nets)	0.001	3.3	3.298409	1.591 drop	0.048	3.298406	1.594 drop	0.048	±3 %	PASS

Load	Group	I (A)	Nominal (V)	V effective (V)	Drop / boost (mV)	%	Worst pin (V)	Worst pin (mV)	Worst %	Tolerance	Result
U903 TXB0102 VCCB	+3.3V (+3 nets)	4e-06	3.3	3.299817	0.183 drop	0.006	3.299817	0.183 drop	0.006	±3 %	PASS
J1903 encoder module	+3.3V (+3 nets)	0.045	3.3	3.299468	0.532 drop	0.016	3.299468	0.532 drop	0.016	±3 %	PASS
U401 74LVC1G97	+3.3V (+3 nets)	1e-05	3.3	3.299983	0.017 drop	0.001	3.299983	0.017 drop	0.001	±3 %	PASS
U601 74LVC1G97	+3.3V (+3 nets)	1e-05	3.3	3.298575	1.425 drop	0.043	3.298575	1.425 drop	0.043	±3 %	PASS
U802 74LVC1G97	+3.3V (+3 nets)	1e-05	3.3	3.298611	1.389 drop	0.042	3.298610	1.390 drop	0.042	±3 %	PASS
U805 74LVC1G97	+3.3V (+3 nets)	1e-05	3.3	3.298835	1.165 drop	0.035	3.298835	1.165 drop	0.035	±3 %	PASS
U1101 74LVC1G97	+3.3V (+3 nets)	1e-05	3.3	3.299723	0.277 drop	0.008	3.299723	0.277 drop	0.008	±3 %	PASS
U1102 74LVC1G97	+3.3V (+3 nets)	1e-05	3.3	3.299605	0.395 drop	0.012	3.299605	0.395 drop	0.012	±3 %	PASS
U1103 74LVC1G97	+3.3V (+3 nets)	1e-05	3.3	3.299550	0.450 drop	0.014	3.299550	0.450 drop	0.014	±3 %	PASS
U1104 74LVC1G97	+3.3V (+3 nets)	1e-05	3.3	3.299656	0.344 drop	0.010	3.299656	0.344 drop	0.010	±3 %	PASS
U1105 74LVC1G97	+3.3V (+3 nets)	1e-05	3.3	3.299497	0.503 drop	0.015	3.299497	0.503 drop	0.015	±3 %	PASS
U1106 74LVC1G97	+3.3V (+3 nets)	1e-05	3.3	3.299446	0.554 drop	0.017	3.299446	0.554 drop	0.017	±3 %	PASS
U1201 74LVC1G97	+3.3V (+3 nets)	1e-05	3.3	3.298525	1.475 drop	0.045	3.298525	1.475 drop	0.045	±3 %	PASS
U1202 74LVC1G97	+3.3V (+3 nets)	1e-05	3.3	3.298491	1.509 drop	0.046	3.298491	1.509 drop	0.046	±3 %	PASS
U1203 74LVC1G97	+3.3V (+3 nets)	1e-05	3.3	3.298457	1.543 drop	0.047	3.298457	1.543 drop	0.047	±3 %	PASS
U1801 74LVC1G97	+3.3V (+3 nets)	1e-05	3.3	3.298496	1.504 drop	0.046	3.298496	1.504 drop	0.046	±3 %	PASS
U1802 74LVC1G97	+3.3V (+3 nets)	1e-05	3.3	3.298406	1.594 drop	0.048	3.298406	1.594 drop	0.048	±3 %	PASS
U1805 74LVC1G97	+3.3V (+3 nets)	1e-05	3.3	3.298502	1.498 drop	0.045	3.298502	1.498 drop	0.045	±3 %	PASS
U1806 74LVC1G97	+3.3V (+3 nets)	1e-05	3.3	3.298412	1.588 drop	0.048	3.298412	1.588 drop	0.048	±3 %	PASS
U1808 74LVC1G97	+3.3V (+3 nets)	1e-05	3.3	3.298433	1.567 drop	0.047	3.298433	1.567 drop	0.047	±3 %	PASS
U1902 74LVC1G97	+3.3V (+3 nets)	1e-05	3.3	3.299080	0.920 drop	0.028	3.299080	0.920 drop	0.028	±3 %	PASS
R1102 LED	+3.3V (+3 nets)	0.0013	3.3	3.299718	0.282 drop	0.009	3.299718	0.282 drop	0.009		
R1103 LED	+3.3V (+3 nets)	0.0013	3.3	3.299601	0.399 drop	0.012	3.299601	0.399 drop	0.012		
R1104 LED	+3.3V (+3 nets)	0.0013	3.3	3.299545	0.455 drop	0.014	3.299545	0.455 drop	0.014		
R1105 LED	+3.3V (+3 nets)	0.0013	3.3	3.299652	0.348 drop	0.011	3.299652	0.348 drop	0.011		
R1106 LED	+3.3V (+3 nets)	0.0013	3.3	3.299493	0.507 drop	0.015	3.299493	0.507 drop	0.015		
R1107 LED	+3.3V (+3 nets)	0.0013	3.3	3.299442	0.558 drop	0.017	3.299442	0.558 drop	0.017		
R1203 LED	+3.3V (+3 nets)	0.0013	3.3	3.298523	1.477 drop	0.045	3.298522	1.478 drop	0.045		

Load	Group	I (A)	Nominal (V)	V effective (V)	Drop / boost (mV)	%	Worst pin (V)	Worst pin (mV)	Worst %	Tolerance	Result
R1204 LED	+3.3V (+3 nets)	0.0013	3.3	3.298488	1.512 drop	0.046	3.298488	1.512 drop	0.046		
R1206 LED	+3.3V (+3 nets)	0.0013	3.3	3.298454	1.546 drop	0.047	3.298454	1.546 drop	0.047		
U902 74LVC1G97	+3.3V_USB	1e-05	3.45	3.449165	0.835 drop	0.024	3.449164	0.836 drop	0.024	±3 %	PASS
U903 TXB0102 VCCA	+3.3V_USB	4e-06	3.45	3.449212	0.788 drop	0.023	3.449212	0.788 drop	0.023	±3 %	PASS
R901 LED	+3.3V_USB	0.0013	3.45	3.449397	0.603 drop	0.017	3.449374	0.626 drop	0.018		
R902 LED	+3.3V_USB	0.0013	3.45	3.449400	0.600 drop	0.017	3.449377	0.623 drop	0.018		
R906 LED	+3.3V_USB	0.0013	3.45	3.449161	0.839 drop	0.024	3.449160	0.840 drop	0.024		

Drop: below the VRM set point. Boost: above it, e.g. near a remote-sensed VRM's output pins. The worst pin is the one furthest from the set point either way. A load passes when every pin is within its tolerance.

VRMs

VRM	Type	Group	V set	Load line (mΩ)	V target	V out	V sense	V gnd sense	I out (A)	V in	I in (A)	P out (W)	P in (W)	Loss (W)	I limit (A)	Result
12V	source	+12V (+2 nets)	12		12.00000	12.00000			3.931			47.17			5	PASS
+3.3V	smps	+3.3V (+3 nets)	3.3		3.30000	3.30348	3.30064	0.00064	0.1377	11.84269	0.04267	0.4548	0.5054	0.05054	1	PASS
+5V_USB	source	+5V_USB (+2 nets)	5		5.00000	5.00000			0.004034			0.02017				
+3.3V_USB	ldo	+3.3V_USB	3.45		3.45000	3.45000			0.003914	4.99898	0.004034	0.0135	0.02017	0.006663	0.1	PASS

DC connections

Part	Group	R (mΩ)	I (A)	P (mW)
R406	+3.3V (+3 nets)	20	0.1377	0.379
R610	+3.3V (+3 nets)	10000	0.007	0.490
FB701	+3.3V (+3 nets)	140	0.0074	0.008
U201	+12V (+2 nets)	28.2	3.931	435.696
R207	+12V (+2 nets)	10	3.931	154.502
FB901	+5V_USB (+2 nets)	140	0.004034	0.002
U904	+5V_USB (+2 nets)	97	0.004034	0.002
FB702	GND (+1 net)	140	0.0074	0.008

Pins

+3.3V (+3 nets): 77 pin(s)

Terminal	Role	Pin	x (mm)	y (mm)	V	Drop / boost (mV)	%	I (mA)	I is
+3.3V	source	L401.2	40.818	55.630	3.303481	3.481 boost	0.105	-137.7	solved
U701 dsPIC VDD	load	U701.15	76.361	88.364	3.300545	0.545 boost	0.017	6.92	equal share
U701 dsPIC VDD	load	U701.28	78.299	92.701	3.300542	0.542 boost	0.016	6.92	equal share
U701 dsPIC VDD	load	U701.36	81.499	92.701	3.300541	0.541 boost	0.016	6.92	equal share
U701 dsPIC VDD	load	U701.41	82.236	90.364	3.300547	0.547 boost	0.017	6.92	equal share
U701 dsPIC VDD	load	U701.46	82.236	88.364	3.300555	0.555 boost	0.017	6.92	equal share

Terminal	Role	Pin	x (mm)	y (mm)	V	Drop / boost (mV)	%	I (mA)	I is
U701 dsPIC AVDD	load	U701.7	79.099	86.826	3.299447	0.553 drop	0.017	7.4	equal share
X601 oscillator	load	X601.4	75.201	95.543	3.230519	69.481 drop	2.105	7	equal share
U1301 MCP6N11	load	U1301.7	98.972	48.024	3.300573	0.573 boost	0.017	1.1	equal share
U1302 MCP6N11	load	U1302.7	113.882	47.991	3.300571	0.571 boost	0.017	1.1	equal share
U1401 MCP6N11	load	U1401.7	74.184	49.558	3.300583	0.583 boost	0.018	1.1	equal share
U1402 MCP6N11	load	U1402.7	87.138	49.558	3.300576	0.576 boost	0.017	1.1	equal share
U1502 MCP6024	load	U1502.4	91.745	59.723	3.300557	0.557 boost	0.017	5.4	equal share
U1601 MCP6024	load	U1601.4	91.745	72.423	3.300562	0.562 boost	0.017	5.4	equal share
U1701 MCP6024	load	U1701.4	117.975	60.300	3.300565	0.565 boost	0.017	5.4	equal share
U1501 TS3A24159	load	U1501.1	95.537	55.204	3.300576	0.576 boost	0.017	0.025	equal share
U1501 TS3A24159	load	U1501.1	95.252	55.204	3.300576	0.576 boost	0.017	0.025	equal share
U1503 MCP4451	load	U1503.15	93.695	63.586	3.300571	0.571 boost	0.017	0.6	equal share
U1702 REF35	load	U1702.4	120.325	59.350	3.300573	0.573 boost	0.017	0.003	equal share
U801 PCA9536	load	U801.8	103.979	94.224	3.300569	0.569 boost	0.017	0.29	equal share
U803 TCA9801	load	U803.1	71.536	96.135	3.300568	0.568 boost	0.017	1.375	equal share
U803 TCA9801	load	U803.8	71.536	100.360	3.300568	0.568 boost	0.017	1.375	equal share
U1807 TCA9801	load	U1807.1	39.088	104.936	3.300582	0.582 boost	0.018	1.375	equal share
U1807 TCA9801	load	U1807.8	39.088	109.161	3.300582	0.582 boost	0.018	1.375	equal share
U1809 TCA9801	load	U1809.1	88.110	104.901	3.300564	0.564 boost	0.017	1.375	equal share
U1809 TCA9801	load	U1809.8	88.110	109.126	3.300564	0.564 boost	0.017	1.375	equal share
U1803 74VHC541	load	U1803.20	56.375	107.320	3.300575	0.575 boost	0.017	1	equal share
U1804 74VHC541	load	U1804.20	105.143	107.320	3.300565	0.565 boost	0.017	1	equal share
U903 TXB0102 VCCB	load	U903.7	65.325	39.856	3.300593	0.593 boost	0.018	0.004	equal share
J1903 encoder module	load	J1903.1	40.183	64.618	3.300633	0.633 boost	0.019	45	equal share
U401 74LVC1G97	load	U401.5	41.915	48.609	3.300634	0.634 boost	0.019	0.01	equal share
U601 74LVC1G97	load	U601.5	75.413	99.670	3.300570	0.570 boost	0.017	0.01	equal share
U802 74LVC1G97	load	U802.1	65.863	98.969	3.300577	0.577 boost	0.017	0.005	equal share
U802 74LVC1G97	load	U802.5	67.538	99.619	3.300574	0.574 boost	0.017	0.005	equal share
U805 74LVC1G97	load	U805.5	86.232	82.652	3.300572	0.572 boost	0.017	0.01	equal share
U1101 74LVC1G97	load	U1101.1	57.354	49.175	3.300599	0.599 boost	0.018	0.005	equal share
U1101 74LVC1G97	load	U1101.5	59.029	49.825	3.300599	0.599 boost	0.018	0.005	equal share
U1102 74LVC1G97	load	U1102.1	57.354	53.747	3.300599	0.599 boost	0.018	0.005	equal share
U1102 74LVC1G97	load	U1102.5	59.029	54.397	3.300599	0.599 boost	0.018	0.005	equal share
U1103 74LVC1G97	load	U1103.1	57.354	56.033	3.300598	0.598 boost	0.018	0.005	equal share
U1103 74LVC1G97	load	U1103.5	59.029	56.683	3.300598	0.598 boost	0.018	0.005	equal share
U1104 74LVC1G97	load	U1104.1	57.375	51.482	3.300599	0.599 boost	0.018	0.005	equal share
U1104 74LVC1G97	load	U1104.5	59.050	52.132	3.300599	0.599 boost	0.018	0.005	equal share
U1105 74LVC1G97	load	U1105.1	57.354	58.319	3.300598	0.598 boost	0.018	0.005	equal share
U1105 74LVC1G97	load	U1105.5	59.029	58.969	3.300598	0.598 boost	0.018	0.005	equal share
U1106 74LVC1G97	load	U1106.1	57.354	60.605	3.300597	0.597 boost	0.018	0.005	equal share
U1106 74LVC1G97	load	U1106.5	59.029	61.255	3.300597	0.597 boost	0.018	0.005	equal share
U1201 74LVC1G97	load	U1201.1	118.593	98.258	3.300568	0.568 boost	0.017	0.005	equal share
U1201 74LVC1G97	load	U1201.5	120.268	98.908	3.300568	0.568 boost	0.017	0.005	equal share
U1202 74LVC1G97	load	U1202.1	118.593	100.544	3.300568	0.568 boost	0.017	0.005	equal share
U1202 74LVC1G97	load	U1202.5	120.268	101.194	3.300568	0.568 boost	0.017	0.005	equal share
U1203 74LVC1G97	load	U1203.1	118.593	102.830	3.300568	0.568 boost	0.017	0.005	equal share
U1203 74LVC1G97	load	U1203.5	120.268	103.480	3.300568	0.568 boost	0.017	0.005	equal share
U1801 74LVC1G97	load	U1801.5	48.057	106.198	3.300584	0.584 boost	0.018	0.01	equal share
U1802 74LVC1G97	load	U1802.5	97.079	106.198	3.300569	0.569 boost	0.017	0.01	equal share
U1805 74LVC1G97	load	U1805.5	43.739	106.198	3.300586	0.586 boost	0.018	0.01	equal share
U1806 74LVC1G97	load	U1806.5	92.761	106.176	3.300569	0.569 boost	0.017	0.01	equal share

Terminal	Role	Pin	x (mm)	y (mm)	V	Drop / boost (mV)	%	I (mA)	I is
U1808 74LVC1G97	load	U1808.1	81.980	106.618	3.300569	0.569 boost	0.017	0.005	equal share
U1808 74LVC1G97	load	U1808.5	83.655	107.268	3.300569	0.569 boost	0.017	0.005	equal share
U1902 74LVC1G97	load	U1902.1	39.599	80.616	3.300605	0.605 boost	0.018	0.005	equal share
U1902 74LVC1G97	load	U1902.5	41.274	81.266	3.300605	0.605 boost	0.018	0.005	equal share
R1102 LED	load	R1102.1	60.478	50.275	3.300595	0.595 boost	0.018	1.3	equal share
R1103 LED	load	R1103.1	60.478	54.847	3.300595	0.595 boost	0.018	1.3	equal share
R1104 LED	load	R1104.1	60.478	57.133	3.300594	0.594 boost	0.018	1.3	equal share
R1105 LED	load	R1105.1	60.499	52.582	3.300595	0.595 boost	0.018	1.3	equal share
R1106 LED	load	R1106.1	60.478	59.419	3.300593	0.593 boost	0.018	1.3	equal share
R1107 LED	load	R1107.1	60.478	61.705	3.300593	0.593 boost	0.018	1.3	equal share
R1203 LED	load	R1203.1	121.717	99.358	3.300565	0.565 boost	0.017	1.3	equal share
R1204 LED	load	R1204.1	121.717	101.644	3.300565	0.565 boost	0.017	1.3	equal share
R1206 LED	load	R1206.1	121.717	103.930	3.300565	0.565 boost	0.017	1.3	equal share
+3.3V (sense)	load	R409.1	37.776	48.667	3.300637	0.637 boost	0.019	0	equal share
R406	DC connection	R406.1	40.824	57.783	3.303456	3.456 boost	0.105	137.7	equal share
R406	DC connection	R406.2	40.824	59.383	3.300702	0.702 boost	0.021	-137.7	equal share
R610	DC connection	R610.1	74.219	96.818	3.300556	0.556 boost	0.017	7	equal share
R610	DC connection	R610.2	74.219	95.918	3.230556	69.444 drop	2.104	-7	equal share
FB701	DC connection	FB701.1	78.283	80.511	3.300565	0.565 boost	0.017	7.4	equal share
FB701	DC connection	FB701.2	78.283	81.411	3.299529	0.471 drop	0.014	-7.4	equal share

+12V (+2 nets): 16 pin(s)

Terminal	Role	Pin	x (mm)	y (mm)	V	Drop / boost (mV)	%	I (mA)	I is
12V	source	J201.1	37.665	33.120	12.000000	0.000 drop	0.000	-3931	solved
J2101	load	J2101.1	71.804	116.942	11.814369	185.631 drop	1.547	486	equal share
J2101	load	J2101.2	71.804	119.482	11.814349	185.651 drop	1.547	486	equal share
J2101	load	J2101.3	69.264	116.942	11.814479	185.521 drop	1.546	486	equal share
J2101	load	J2101.4	69.264	119.482	11.814477	185.523 drop	1.546	486	equal share
J2102	load	J2102.1	120.754	116.942	11.804045	195.955 drop	1.633	486	equal share
J2102	load	J2102.2	120.754	119.482	11.804031	195.969 drop	1.633	486	equal share
J2102	load	J2102.3	118.214	116.942	11.804076	195.924 drop	1.633	486	equal share
J2102	load	J2102.4	118.214	119.482	11.804071	195.929 drop	1.633	486	equal share
+3.3V (input)	load	U402.10	35.043	52.237	11.842595	157.405 drop	1.312	14.22	equal share
+3.3V (input)	load	U402.11	35.043	52.737	11.842744	157.256 drop	1.310	14.22	equal share
+3.3V (input)	load	U402.12	35.043	53.237	11.842744	157.256 drop	1.310	14.22	equal share
U201	DC connection	U201.5	35.615	38.964	11.998023	1.977 drop	0.016	3931	equal share
U201	DC connection	U201.6	36.115	38.964	11.887178	112.822 drop	0.940	-3931	equal share
R207	DC connection	R207.1	35.865	41.466	11.885882	114.118 drop	0.951	3931	equal share
R207	DC connection	R207.2	35.865	43.066	11.846575	153.425 drop	1.279	-3931	equal share

+3.3V_USB: 8 pin(s)

Terminal	Role	Pin	x (mm)	y (mm)	V	Drop / boost (mV)	%	I (mA)	I is
+3.3V_USB	source	U901.5	57.975	36.506	3.450000	0.000 drop	0.000	-2.271	solved
+3.3V_USB	source	U901.6	57.475	36.506	3.450000	0.000 drop	0.000	-1.643	solved
U902 74LVC1G97	load	U902.1	57.430	46.814	3.449983	0.017 drop	0.000	0.005	equal share
U902 74LVC1G97	load	U902.5	59.105	47.464	3.449983	0.017 drop	0.000	0.005	equal share
U903 TXB0102 VCCA	load	U903.3	64.825	37.056	3.449988	0.012 drop	0.000	0.004	equal share
R901 LED	load	R901.1	56.439	42.578	3.449974	0.026 drop	0.001	1.3	equal share
R902 LED	load	R902.1	57.709	42.578	3.449977	0.023 drop	0.001	1.3	equal share
R906 LED	load	R906.1	60.477	47.948	3.449979	0.021 drop	0.001	1.3	equal share

Terminal	Role	Pin	x (mm)	y (mm)	V	Drop / boost (mV)	%	I (mA)	I is

+5V_USB (+2 nets): 11 pin(s)

Terminal	Role	Pin	x (mm)	y (mm)	V	Drop / boost (mV)	%	I (mA)	I is
+5V_USB	source	J901.A4	61.075	29.822	5.000000	0.000 drop	0.000	-0	solved
+5V_USB	source	J901.A9	56.375	29.822	5.000000	0.000 drop	0.000	-2.017	solved
+5V_USB	source	J901.B4	56.375	29.822	5.000000	0.000 drop	0.000	-2.017	solved
+5V_USB	source	J901.B9	61.075	29.822	5.000000	0.000 drop	0.000	-0	solved
+3.3V_USB (input)	load	U901.7	56.775	37.206	4.998985	1.015 drop	0.020	2.017	equal share
+3.3V_USB (input)	load	U901.8	56.775	37.706	4.998984	1.016 drop	0.020	2.017	equal share
FB901	DC connection	FB901.2	54.857	32.156	4.999989	0.011 drop	0.000	4.034	equal share
FB901	DC connection	FB901.1	53.957	32.156	4.999424	0.576 drop	0.012	-4.034	equal share
U904	DC connection	U904.1	53.307	34.326	4.999412	0.588 drop	0.012	2.017	equal share
U904	DC connection	U904.3	53.307	32.426	4.999412	0.588 drop	0.012	2.017	equal share
U904	DC connection	U904.4	55.507	32.426	4.999020	0.980 drop	0.020	-4.034	equal share

GND (+1 net): 162 pin(s)

Terminal	Role	Pin	x (mm)	y (mm)	V	Drop / boost (mV)	%	I (mA)	I is
12V (gnd)	source	J201.2	47.165	25.820	0.000000	0.000 drop		450.7	solved
12V (gnd)	source	J201.2	37.265	25.820	0.000000	0.000 drop		323.6	solved
12V (gnd)	source	J201.2	46.265	33.120	0.000000	0.000 drop		1436	solved
12V (gnd)	source	J201.3	41.065	33.120	0.000000	0.000 drop		890.4	solved
+5V_USB (gnd)	source	J901.A1	61.825	29.822	0.000000	0.000 drop		253.4	solved
+5V_USB (gnd)	source	J901.B1	55.625	29.822	0.000000	0.000 drop		163.8	solved
+5V_USB (gnd)	source	J901.A12	55.625	29.822	0.000000	0.000 drop		163.8	solved
+5V_USB (gnd)	source	J901.B12	61.825	29.822	0.000000	0.000 drop		253.4	solved
J2101 (return)	load	J2101.5	66.724	116.942	0.002330	2.330 boost		-121.5	equal share
J2101 (return)	load	J2101.6	66.724	119.482	0.002343	2.343 boost		-121.5	equal share
J2101 (return)	load	J2101.7	64.184	116.942	0.002333	2.333 boost		-121.5	equal share
J2101 (return)	load	J2101.8	64.184	119.482	0.002350	2.350 boost		-121.5	equal share
J2101 (return)	load	J2101.10	61.644	119.482	0.002365	2.365 boost		-121.5	equal share
J2101 (return)	load	J2101.12	59.104	119.482	0.002373	2.373 boost		-121.5	equal share
J2101 (return)	load	J2101.14	56.564	119.482	0.002376	2.376 boost		-121.5	equal share
J2101 (return)	load	J2101.16	54.024	119.482	0.002376	2.376 boost		-121.5	equal share
J2101 (return)	load	J2101.18	51.484	119.482	0.002374	2.374 boost		-121.5	equal share
J2101 (return)	load	J2101.20	48.944	119.482	0.002369	2.369 boost		-121.5	equal share
J2101 (return)	load	J2101.22	46.404	119.482	0.002362	2.362 boost		-121.5	equal share
J2101 (return)	load	J2101.24	43.864	119.482	0.002353	2.353 boost		-121.5	equal share
J2101 (return)	load	J2101.26	41.324	119.482	0.002341	2.341 boost		-121.5	equal share
J2101 (return)	load	J2101.28	38.784	119.482	0.002325	2.325 boost		-121.5	equal share
J2101 (return)	load	J2101.29	36.244	116.942	0.002277	2.277 boost		-121.5	equal share
J2101 (return)	load	J2101.30	36.244	119.482	0.002304	2.304 boost		-121.5	equal share
J2102 (return)	load	J2102.5	115.674	116.942	0.002353	2.353 boost		-121.5	equal share
J2102 (return)	load	J2102.6	115.674	119.482	0.002365	2.365 boost		-121.5	equal share
J2102 (return)	load	J2102.7	113.134	116.942	0.002358	2.358 boost		-121.5	equal share
J2102 (return)	load	J2102.8	113.134	119.482	0.002374	2.374 boost		-121.5	equal share
J2102 (return)	load	J2102.10	110.594	119.482	0.002394	2.394 boost		-121.5	equal share
J2102 (return)	load	J2102.12	108.054	119.482	0.002406	2.406 boost		-121.5	equal share
J2102 (return)	load	J2102.14	105.514	119.482	0.002413	2.413 boost		-121.5	equal share
J2102 (return)	load	J2102.16	102.974	119.482	0.002416	2.416 boost		-121.5	equal share
J2102 (return)	load	J2102.18	100.434	119.482	0.002416	2.416 boost		-121.5	equal share
J2102 (return)	load	J2102.20	97.894	119.482	0.002413	2.413 boost		-121.5	equal share

Terminal	Role	Pin	x (mm)	y (mm)	V	Drop / boost (mV)	%	I (mA)	I is
J2102 (return)	load	J2102.22	95.354	119.482	0.002406	2.406 boost		-121.5	equal share
J2102 (return)	load	J2102.24	92.814	119.482	0.002396	2.396 boost		-121.5	equal share
J2102 (return)	load	J2102.26	90.274	119.482	0.002383	2.383 boost		-121.5	equal share
J2102 (return)	load	J2102.28	87.734	119.482	0.002365	2.365 boost		-121.5	equal share
J2102 (return)	load	J2102.29	85.194	116.942	0.002316	2.316 boost		-121.5	equal share
J2102 (return)	load	J2102.30	85.194	119.482	0.002342	2.342 boost		-121.5	equal share
U701 dsPIC VDD (return)	load	U701.14	76.361	87.964	0.001791	1.791 boost		-5.767	equal share
U701 dsPIC VDD (return)	load	U701.27	77.899	92.701	0.001931	1.931 boost		-5.767	equal share
U701 dsPIC VDD (return)	load	U701.35	81.099	92.701	0.001944	1.944 boost		-5.767	equal share
U701 dsPIC VDD (return)	load	U701.43	82.236	89.564	0.001856	1.856 boost		-5.767	equal share
U701 dsPIC VDD (return)	load	U701.45	82.236	88.764	0.001851	1.851 boost		-5.767	equal share
U701 dsPIC VDD (return)	load	U701.49	79.299	89.764	0.001835	1.835 boost		-5.767	equal share
U701 dsPIC AVDD (return)	load	U701.6	79.499	86.826	0.002795	2.795 boost		-7.4	equal share
X601 oscillator (return)	load	X601.2	77.301	97.193	0.001975	1.975 boost		-7	equal share
U1301 MCP6N11 (return)	load	U1301.4	99.972	45.024	0.001394	1.394 boost		-0.55	equal share
U1301 MCP6N11 (return)	load	U1301.9	99.222	46.524	0.001393	1.393 boost		-0.55	equal share
U1302 MCP6N11 (return)	load	U1302.4	114.882	44.991	0.001477	1.477 boost		-0.55	equal share
U1302 MCP6N11 (return)	load	U1302.9	114.132	46.491	0.001477	1.477 boost		-0.55	equal share
U1401 MCP6N11 (return)	load	U1401.4	75.184	46.558	0.001120	1.120 boost		-0.55	equal share
U1401 MCP6N11 (return)	load	U1401.9	74.434	48.058	0.001118	1.118 boost		-0.55	equal share
U1402 MCP6N11 (return)	load	U1402.4	88.138	46.558	0.001292	1.292 boost		-0.55	equal share
U1402 MCP6N11 (return)	load	U1402.9	87.388	48.058	0.001289	1.289 boost		-0.55	equal share
U1502 MCP6024 (return)	load	U1502.11	91.745	65.448	0.001515	1.515 boost		-5.4	equal share
U1601 MCP6024 (return)	load	U1601.11	91.745	78.148	0.001695	1.695 boost		-5.4	equal share
U1701 MCP6024 (return)	load	U1701.11	112.250	60.300	0.001591	1.591 boost		-5.4	equal share
U1501 TS3A24159 (return)	load	U1501.6	92.302	53.204	0.001379	1.379 boost		-0.003333	equal share
U1501 TS3A24159 (return)	load	U1501.6	92.017	53.204	0.001379	1.379 boost		-0.003333	equal share
U1501 TS3A24159 (return)	load	U1501.11	94.227	54.839	0.001414	1.414 boost		-0.003333	equal share
U1501 TS3A24159 (return)	load	U1501.11	94.227	53.569	0.001414	1.414 boost		-0.003333	equal share
U1501 TS3A24159 (return)	load	U1501.11	94.027	55.834	0.001414	1.414 boost		-0.003333	equal share
U1501 TS3A24159 (return)	load	U1501.11	94.027	55.754	0.001414	1.414 boost		-0.003333	equal share
U1501 TS3A24159 (return)	load	U1501.11	94.027	52.654	0.001414	1.414 boost		-0.003333	equal share
U1501 TS3A24159 (return)	load	U1501.11	94.027	52.574	0.001414	1.414 boost		-0.003333	equal share
U1501 TS3A24159 (return)	load	U1501.11	93.777	54.204	0.001414	1.414 boost		-0.003333	equal share
U1501 TS3A24159 (return)	load	U1501.11	93.527	55.834	0.001414	1.414 boost		-0.003333	equal share
U1501 TS3A24159 (return)	load	U1501.11	93.527	55.754	0.001414	1.414 boost		-0.003333	equal share
U1501 TS3A24159 (return)	load	U1501.11	93.527	52.654	0.001414	1.414 boost		-0.003333	equal share
U1501 TS3A24159 (return)	load	U1501.11	93.527	52.574	0.001414	1.414 boost		-0.003333	equal share
U1501 TS3A24159 (return)	load	U1501.11	93.327	54.839	0.001414	1.414 boost		-0.003333	equal share
U1501 TS3A24159 (return)	load	U1501.11	93.327	53.569	0.001414	1.414 boost		-0.003333	equal share
U1503 MCP4451 (return)	load	U1503.2	89.795	63.086	0.001481	1.481 boost		-0.06	equal share
U1503 MCP4451 (return)	load	U1503.5	89.795	61.586	0.001458	1.458 boost		-0.06	equal share
U1503 MCP4451 (return)	load	U1503.6	90.745	60.636	0.001491	1.491 boost		-0.06	equal share
U1503 MCP4451 (return)	load	U1503.7	91.245	60.636	0.001491	1.491 boost		-0.06	equal share
U1503 MCP4451 (return)	load	U1503.8	91.745	60.636	0.001491	1.491 boost		-0.06	equal share
U1503 MCP4451 (return)	load	U1503.9	92.245	60.636	0.001491	1.491 boost		-0.06	equal share
U1503 MCP4451 (return)	load	U1503.10	92.745	60.636	0.001491	1.491 boost		-0.06	equal share
U1503 MCP4451 (return)	load	U1503.11	93.695	61.586	0.001497	1.497 boost		-0.06	equal share
U1503 MCP4451 (return)	load	U1503.14	93.695	63.086	0.001530	1.530 boost		-0.06	equal share
U1503 MCP4451 (return)	load	U1503.21	91.745	62.586	0.001491	1.491 boost		-0.06	equal share
U1702 REF35 (return)	load	U1702.1	122.600	61.250	0.001607	1.607 boost		-0.0015	equal share

Terminal	Role	Pin	x (mm)	y (mm)	V	Drop / boost (mV)	%	I (mA)	I is
U1702 REF35 (return)	load	U1702.2	122.600	60.300	0.001607	1.607 boost		-0.0015	equal share
U801 PCA9536 (return)	load	U801.4	99.679	96.174	0.001982	1.982 boost		-0.29	equal share
U803 TCA9801 (return)	load	U803.4	69.586	96.135	0.001947	1.947 boost		-2.75	equal share
U1807 TCA9801 (return)	load	U1807.4	37.138	104.936	0.002073	2.073 boost		-2.75	equal share
U1809 TCA9801 (return)	load	U1809.4	86.160	104.901	0.002142	2.142 boost		-2.75	equal share
U1803 74VHC541 (return)	load	U1803.10	52.000	106.820	0.002080	2.080 boost		-0.5	equal share
U1803 74VHC541 (return)	load	U1803.21	54.188	107.070	0.002085	2.085 boost		-0.5	equal share
U1804 74VHC541 (return)	load	U1804.10	100.768	106.820	0.002154	2.154 boost		-0.5	equal share
U1804 74VHC541 (return)	load	U1804.21	102.956	107.070	0.002159	2.159 boost		-0.5	equal share
U903 TXB0102 VCCB (return)	load	U903.2	65.325	37.056	0.000775	0.775 boost		-0.004	equal share
J1903 encoder module (return)	load	J1903.3	45.263	64.618	0.001166	1.166 boost		-45	equal share
U401 74LVC1G97 (return)	load	U401.1	40.240	47.959	0.000651	0.651 boost		-0.005	equal share
U401 74LVC1G97 (return)	load	U401.2	40.240	48.609	0.000651	0.651 boost		-0.005	equal share
U601 74LVC1G97 (return)	load	U601.2	77.088	99.670	0.001995	1.995 boost		-0.003333	equal share
U601 74LVC1G97 (return)	load	U601.3	77.088	99.020	0.001995	1.995 boost		-0.003333	equal share
U601 74LVC1G97 (return)	load	U601.6	75.413	100.320	0.001995	1.995 boost		-0.003333	equal share
U802 74LVC1G97 (return)	load	U802.2	65.863	99.619	0.001965	1.965 boost		-0.005	equal share
U802 74LVC1G97 (return)	load	U802.3	65.863	100.269	0.001965	1.965 boost		-0.005	equal share
U805 74LVC1G97 (return)	load	U805.2	84.557	82.652	0.001736	1.736 boost		-0.003333	equal share
U805 74LVC1G97 (return)	load	U805.3	84.557	83.302	0.001736	1.736 boost		-0.003333	equal share
U805 74LVC1G97 (return)	load	U805.6	86.232	82.002	0.001736	1.736 boost		-0.003333	equal share
U1101 74LVC1G97 (return)	load	U1101.2	57.354	49.825	0.000877	0.877 boost		-0.01	equal share
U1102 74LVC1G97 (return)	load	U1102.2	57.354	54.397	0.000994	0.994 boost		-0.01	equal share
U1103 74LVC1G97 (return)	load	U1103.2	57.354	56.683	0.001049	1.049 boost		-0.01	equal share
U1104 74LVC1G97 (return)	load	U1104.2	57.375	52.132	0.000943	0.943 boost		-0.01	equal share
U1105 74LVC1G97 (return)	load	U1105.2	57.354	58.969	0.001100	1.100 boost		-0.01	equal share
U1106 74LVC1G97 (return)	load	U1106.2	57.354	61.255	0.001151	1.151 boost		-0.01	equal share
U1201 74LVC1G97 (return)	load	U1201.2	118.593	98.908	0.002042	2.042 boost		-0.005	equal share
U1201 74LVC1G97 (return)	load	U1201.3	118.593	99.558	0.002043	2.043 boost		-0.005	equal share
U1202 74LVC1G97 (return)	load	U1202.2	118.593	101.194	0.002077	2.077 boost		-0.005	equal share
U1202 74LVC1G97 (return)	load	U1202.3	118.593	101.844	0.002077	2.077 boost		-0.005	equal share
U1203 74LVC1G97 (return)	load	U1203.2	118.593	103.480	0.002111	2.111 boost		-0.005	equal share
U1203 74LVC1G97 (return)	load	U1203.3	118.593	104.130	0.002111	2.111 boost		-0.005	equal share
U1801 74LVC1G97 (return)	load	U1801.2	48.057	107.873	0.002088	2.088 boost		-0.005	equal share
U1801 74LVC1G97 (return)	load	U1801.3	48.707	107.873	0.002088	2.088 boost		-0.005	equal share
U1802 74LVC1G97 (return)	load	U1802.2	97.079	107.873	0.002163	2.163 boost		-0.005	equal share
U1802 74LVC1G97 (return)	load	U1802.3	97.729	107.873	0.002163	2.163 boost		-0.005	equal share
U1805 74LVC1G97 (return)	load	U1805.2	43.739	107.873	0.002083	2.083 boost		-0.005	equal share
U1805 74LVC1G97 (return)	load	U1805.3	44.389	107.873	0.002083	2.083 boost		-0.005	equal share
U1806 74LVC1G97 (return)	load	U1806.2	92.761	107.851	0.002157	2.157 boost		-0.005	equal share
U1806 74LVC1G97 (return)	load	U1806.3	93.411	107.851	0.002157	2.157 boost		-0.005	equal share
U1808 74LVC1G97 (return)	load	U1808.2	81.980	107.268	0.002137	2.137 boost		-0.005	equal share
U1808 74LVC1G97 (return)	load	U1808.3	81.980	107.918	0.002137	2.137 boost		-0.005	equal share
U1902 74LVC1G97 (return)	load	U1902.2	39.599	81.266	0.001524	1.524 boost		-0.005	equal share
U1902 74LVC1G97 (return)	load	U1902.3	39.599	81.916	0.001524	1.524 boost		-0.005	equal share
R1102 LED (return)	load	U1101.2	57.354	49.825	0.000877	0.877 boost		-1.3	equal share
R1103 LED (return)	load	U1102.2	57.354	54.397	0.000994	0.994 boost		-1.3	equal share
R1104 LED (return)	load	U1103.2	57.354	56.683	0.001049	1.049 boost		-1.3	equal share
R1105 LED (return)	load	U1104.2	57.375	52.132	0.000943	0.943 boost		-1.3	equal share
R1106 LED (return)	load	U1105.2	57.354	58.969	0.001100	1.100 boost		-1.3	equal share
R1107 LED (return)	load	U1106.2	57.354	61.255	0.001151	1.151 boost		-1.3	equal share

Terminal	Role	Pin	x (mm)	y (mm)	V	Drop / boost (mV)	%	I (mA)	I is
R1203 LED (return)	load	U1201.2	118.593	98.908	0.002042	2.042 boost		-0.65	equal share
R1203 LED (return)	load	U1201.3	118.593	99.558	0.002043	2.043 boost		-0.65	equal share
R1204 LED (return)	load	U1202.2	118.593	101.194	0.002077	2.077 boost		-0.65	equal share
R1204 LED (return)	load	U1202.3	118.593	101.844	0.002077	2.077 boost		-0.65	equal share
R1206 LED (return)	load	U1203.2	118.593	103.480	0.002111	2.111 boost		-0.65	equal share
R1206 LED (return)	load	U1203.3	118.593	104.130	0.002111	2.111 boost		-0.65	equal share
U902 74LVC1G97 (return)	load	U902.2	57.430	47.464	0.000818	0.818 boost		-0.005	equal share
U902 74LVC1G97 (return)	load	U902.3	57.430	48.114	0.000819	0.819 boost		-0.005	equal share
U903 TXB0102 VCCA (return)	load	U903.2	65.325	37.056	0.000775	0.775 boost		-0.004	equal share
R901 LED (return)	load	U901.2	59.475	36.506	0.000553	0.553 boost		-0.65	equal share
R901 LED (return)	load	U901.25	58.725	38.456	0.000600	0.600 boost		-0.65	equal share
R902 LED (return)	load	U901.2	59.475	36.506	0.000553	0.553 boost		-0.65	equal share
R902 LED (return)	load	U901.25	58.725	38.456	0.000600	0.600 boost		-0.65	equal share
R906 LED (return)	load	U902.2	57.430	47.464	0.000818	0.818 boost		-0.65	equal share
R906 LED (return)	load	U902.3	57.430	48.114	0.000819	0.819 boost		-0.65	equal share
+3.3V (gnd)	load	U402.6	36.756	51.025	0.000728	0.728 boost		15.84	equal share
+3.3V (gnd)	load	U402.7	36.256	51.025	0.000728	0.728 boost		15.84	equal share
+3.3V (gnd)	load	U402.8	35.756	51.025	0.000727	0.727 boost		15.84	equal share
+3.3V (gnd)	load	U402.15	36.756	53.950	0.000728	0.728 boost		15.84	equal share
+3.3V (gnd)	load	U402.16	37.256	53.950	0.000727	0.727 boost		15.84	equal share
+3.3V (gnd)	load	U402.17	36.506	52.487	0.000734	0.734 boost		15.84	equal share
+3.3V_USB (gnd)	load	U901.2	59.475	36.506	0.000553	0.553 boost		-0.06	equal share
+3.3V_USB (gnd)	load	U901.25	58.725	38.456	0.000600	0.600 boost		-0.06	equal share
+3.3V (gnd sense)	load	R410.2	36.760	48.667	0.000637	0.637 boost		0	equal share
FB702	DC connection	FB702.2	80.289	81.400	0.002712	2.712 boost		7.4	equal share
FB702	DC connection	FB702.1	80.289	80.500	0.001676	1.676 boost		-7.4	equal share

Every pin each group was solved with: VRM outputs, phases, inputs and sense pins, load supply and ground pins and the pins of DC connections. I is the current drawn from the copper at the pin; negative is delivered into it (a VRM output, or a DC connection's far side). VRM output pin currents are solved; a load's current is shared equally by its pins, as the solver applies it, and a DC connection's by the pins on each side. Drop and boost are from the VRM set point (0 V for a ground).

Losses

Group	F.Cu (mW)	In1.Cu (mW)	In2.Cu (mW)	In3.Cu (mW)	In4.Cu (mW)	In5.Cu (mW)	In6.Cu (mW)	B.Cu (mW)	Vias (mW)	DC parts (mW)	Total (mW)
+3.3V (+3 nets)	0.005	0.000	0.013	0.001	0.000	0.000	0.000	0.000	0.001	0.877	0.897
+12V (+2 nets)	7.433	0.000	2.326	143.287	0.000	0.000	0.000	2.338	2.816	590.198	748.399
+3.3V_USB	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
+5V_USB (+2 nets)	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.004	0.004
GND (+1 net)	0.131	3.111	0.000	0.000	2.982	0.000	2.970	0.013	0.135	0.008	9.349

Highest via currents

Group	Via / pad	x (mm)	y (mm)	Drill (mm)	I (mA)	P (mW)	Result
+12V (+2 nets)	via	35.611	37.948	0.200	803.94	0.4143	
+12V (+2 nets)	via	36.119	39.980	0.200	707.70	0.2304	
GND (+1 net)	via	61.824	30.734	0.200	506.74	0.0809	
+12V (+2 nets)	via	35.611	38.456	0.200	455.73	0.2194	

Group	Via / pad	x (mm)	y (mm)	Drill (mm)	I (mA)	P (mW)	Result
+12V (+2 nets)	via	36.906	41.250	0.200	439.60	0.1122	
+12V (+2 nets)	via	36.398	41.250	0.200	383.90	0.0950	
+12V (+2 nets)	via	36.881	43.586	0.200	376.46	0.1186	
+12V (+2 nets)	via	36.119	39.472	0.200	363.78	0.1011	
+12V (+2 nets)	via	34.849	43.586	0.200	351.26	0.1033	
+12V (+2 nets)	via	36.881	44.602	0.200	340.68	0.0968	
+12V (+2 nets)	via	35.890	41.250	0.200	328.23	0.0782	
GND (+1 net)	via	55.626	30.759	0.200	327.60	0.0374	
+12V (+2 nets)	via	35.611	38.964	0.200	324.24	0.1430	
+12V (+2 nets)	via	34.849	44.602	0.200	312.44	0.0815	
+12V (+2 nets)	via	35.382	41.250	0.200	267.45	0.0631	
+12V (+2 nets)	via	36.881	44.094	0.200	267.38	0.0597	
+12V (+2 nets)	via	36.373	43.586	0.200	264.42	0.0586	
+12V (+2 nets)	via	35.611	39.472	0.200	263.91	0.1068	
+12V (+2 nets)	via	35.357	43.586	0.200	254.38	0.0542	
+12V (+2 nets)	via	34.849	44.094	0.200	245.24	0.0503	
+12V (+2 nets)	via	36.119	38.964	0.200	240.45	0.0614	
+12V (+2 nets)	via	35.611	39.980	0.200	236.80	0.0910	
+12V (+2 nets)	via	34.874	41.250	0.200	235.66	0.0561	
+12V (+2 nets)	via	35.865	43.586	0.200	232.55	0.0453	
+12V (+2 nets)	via	36.373	44.602	0.200	230.03	0.0442	

Electro-thermal

0.816 W into the board, load power not included; peak **60.2 °C** at (35.5, 39.1) mm on F.Cu. Ambient 25 °C, airflow 0 m/s, 5 iteration(s).

Heat by source: DC connections 0.591 W, copper 0.165 W, VRM conversion 0.0572 W, vias 0.00295 W.

Component power dissipation

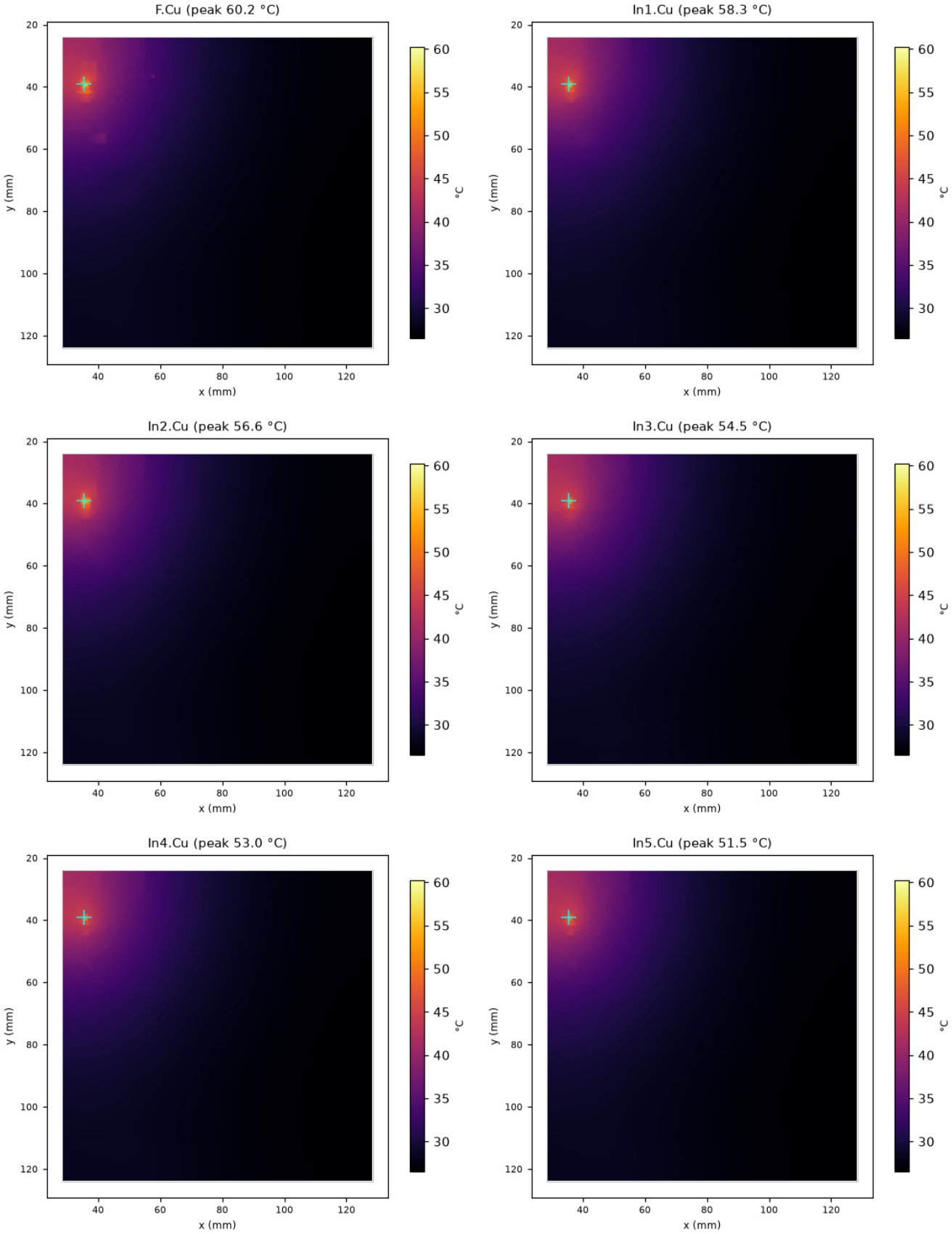
Part	Kind	Power (W)
R406	DC connection	0.0003792
R610	DC connection	0.00049
FB701	DC connection	7.666e-06
U201	DC connection	0.4357
R207	DC connection	0.1545
FB901	DC connection	2.278e-06
U904	DC connection	1.578e-06
FB702	DC connection	7.666e-06

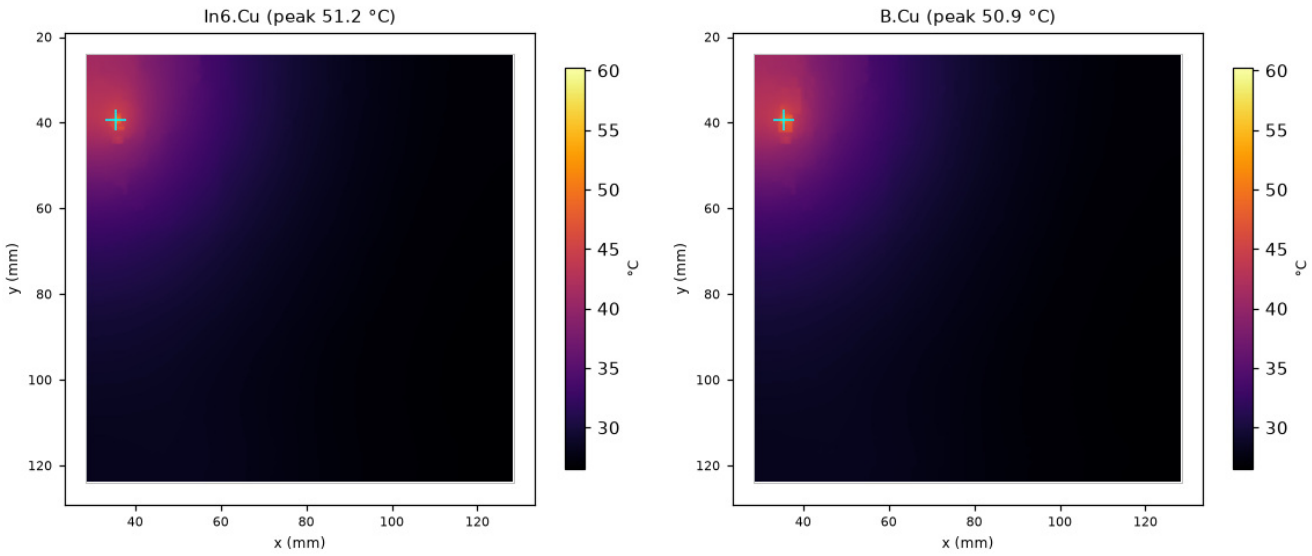
Heat put into the board at each part's pins. A DC connection dissipates I^2R. Load power is not included (Thermal settings tab).

Copper temperature by layer

Layer	Peak (°C)	Rise (°C)	x (mm)	y (mm)	Group	Result
F.Cu	60.1	35.1	35.5	39.1	+12V (+2 nets)	**FAIL**
In1.Cu	54.7	29.7	35.4	39.2	GND (+1 net)	**FAIL**
In2.Cu	56.6	31.6	35.5	39.1	+12V (+2 nets)	**FAIL**
In3.Cu	51.8	26.8	35.4	39.2	+3.3V (+3 nets)	**FAIL**
In4.Cu	50.8	25.8	35.4	39.2	GND (+1 net)	**FAIL**
In5.Cu	42.7	17.7	37.0	34.5	+12V (+2 nets)	**FAIL**
In6.Cu	49.8	24.8	35.4	39.2	GND (+1 net)	**FAIL**
B.Cu	50.9	25.9	35.5	39.5	+12V (+2 nets)	**FAIL**

The hottest solved copper on each layer. Limits: copper at most 40 °C, at most 20 °C above ambient; overall **FAIL**.





Laminate (thermal conductivity)

Stackup material	Thickness (mm)	Taken as	In-plane k (W/m·K)	Through-plane k (W/m·K)
Top silkscreen	0.015	silkscreen, where it prints	0.2	0.2
Top solder mask	0.010	solder mask	0.25	0.25
3313 RC57%	0.099	FR4 prepreg	0.6	0.25
FR4 core	0.100	FR4 core	0.8	0.3
2116 RC54% x3	0.357	FR4 prepreg	0.6	0.25
FR4 core	0.300	FR4 core	0.8	0.3
2116 RC54% x3	0.357	FR4 prepreg	0.6	0.25
FR4 core	0.100	FR4 core	0.8	0.3
3313 RC57%	0.099	FR4 prepreg	0.6	0.25
Bottom solder mask	0.010	solder mask	0.25	0.25
Bottom silkscreen	0.015	silkscreen, where it prints	0.2	0.2

Iteration	Heat (W)	Peak (°C)	Max change (°C)
1	0.8117	60.11	35.112
2	0.8158	60.18	0.106
3	0.8159	60.18	0.000
4	0.8159	60.18	0.000
5	0.8159	60.18	0.000

+3.3V (+3 nets)

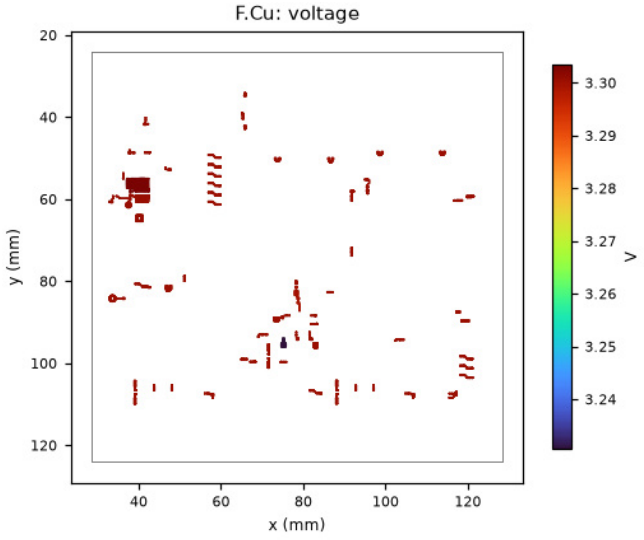
driven by +3.3V; nets: +3.3V, +3.3VA_DSP, Net-(C406-Pad1), Net-(X601-Vdd).

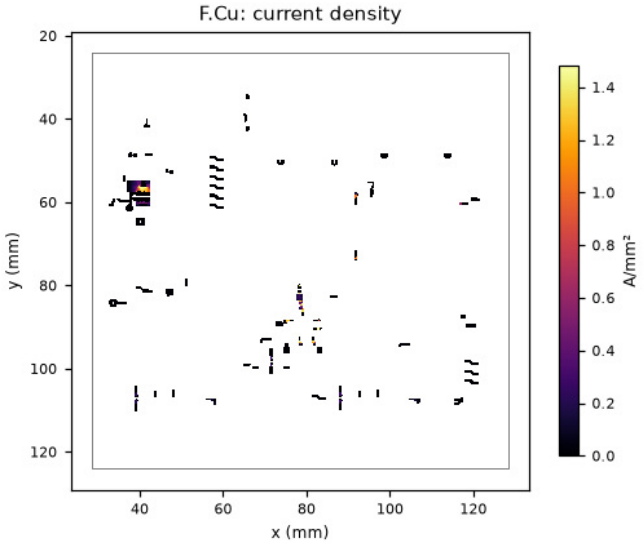
Interactive 3D view: Audio_Visualiser_DSP_kipi_report_2026-10-18_211614_3d.html

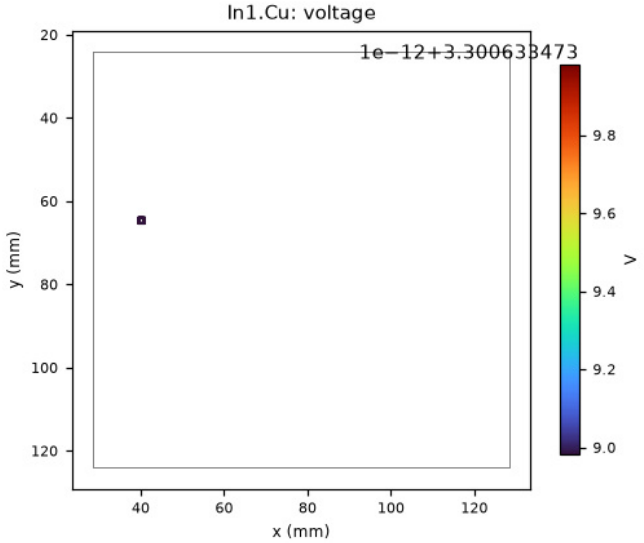
Layer	Copper (mm²)	V min	V max	Max drop (mV)	Max boost (mV)	|J| p99.5 (A/mm²)	|J| max	Loss (mW)
F.Cu	79.1	3.230519	3.303481	69.481	3.481	1.48	6.56	0.005
In1.Cu	2.1	3.300633	3.300633	0.000	0.633	1.76e-09	2.68e-09	0.000
In2.Cu	8776.0	3.300563	3.300697	0.000	0.697	1.27	3.82	0.013
In3.Cu	6469.0	3.300564	3.300696	0.000	0.696	0.289	0.985	0.001
In4.Cu	2.1	3.300633	3.300633	0.000	0.633	1.76e-09	2.68e-09	0.000
In5.Cu	2.1	3.300633	3.300633	0.000	0.633	1.76e-09	2.68e-09	0.000
In6.Cu	2.1	3.300633	3.300633	0.000	0.633	1.76e-09	2.68e-09	0.000
B.Cu	42.4	3.230536	3.303470	69.464	3.470	0.71	1.19	0.000

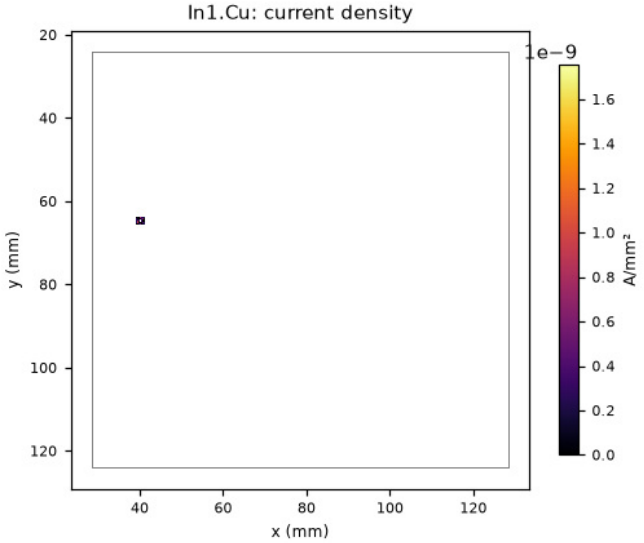
Layer	Copper (mm²)	V min	V max	Max drop (mV)	Max boost (mV)	|J| p99.5 (A/mm²)	|J| max	Loss (mW)

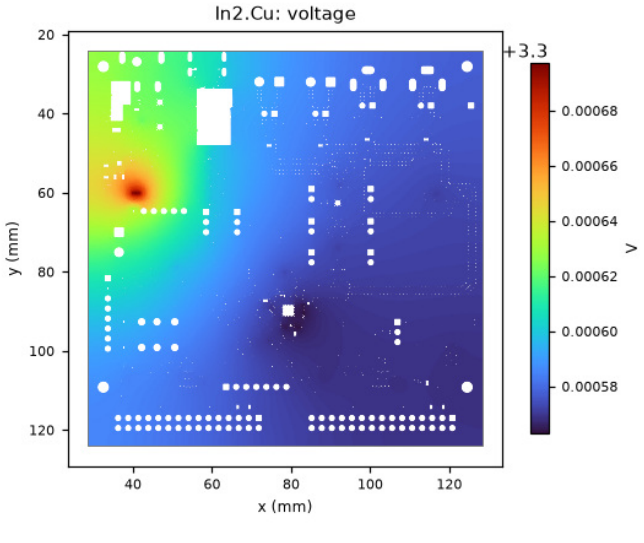
Drop and boost are measured from the 3.3 V set point.

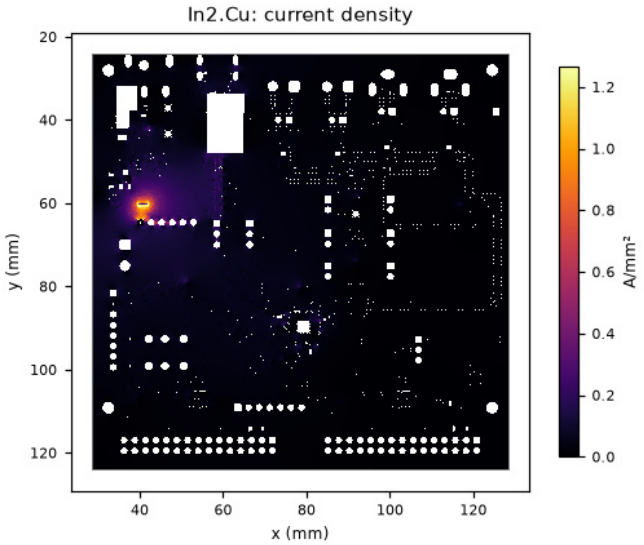


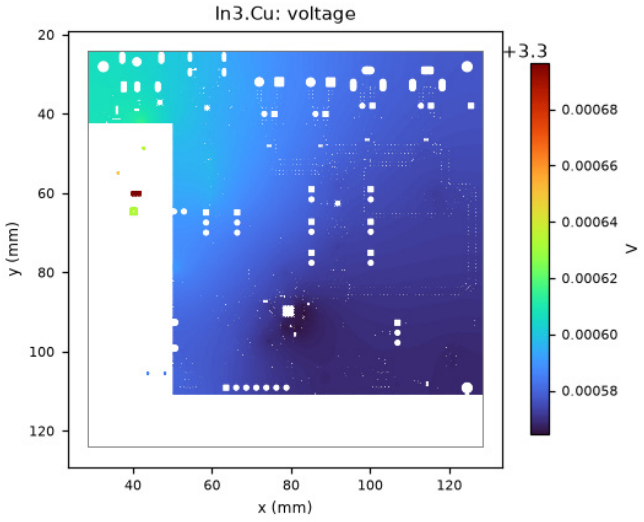


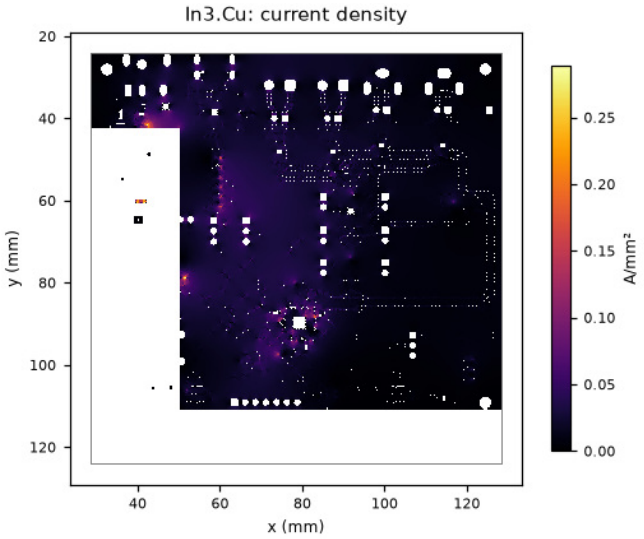


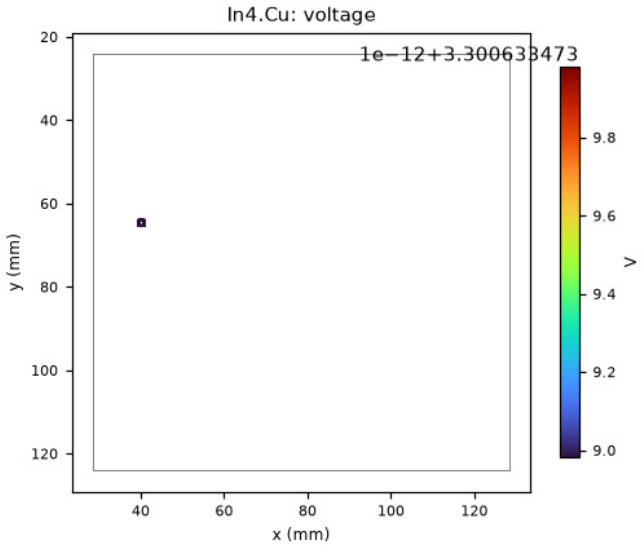


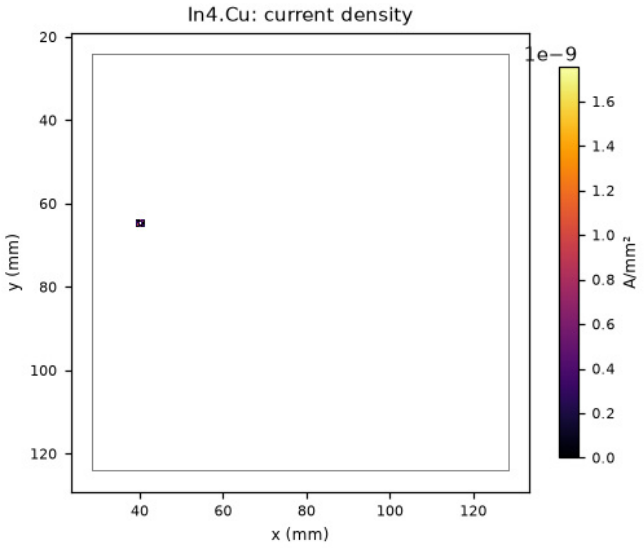


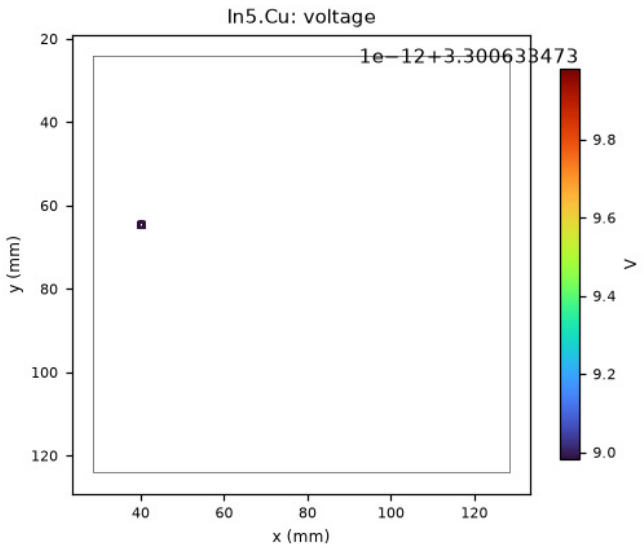


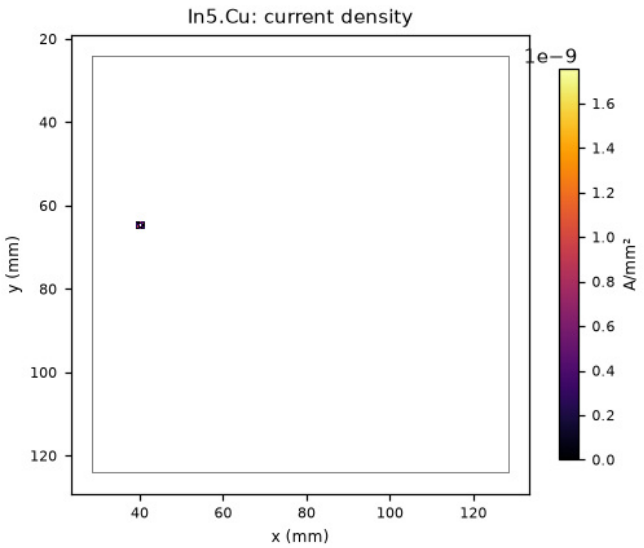

In3.Cu: voltage
+3.3
0.00068
0.00066
0.00064
0.00062
0.00060
0.00058
V
y (mm)
x (mm)

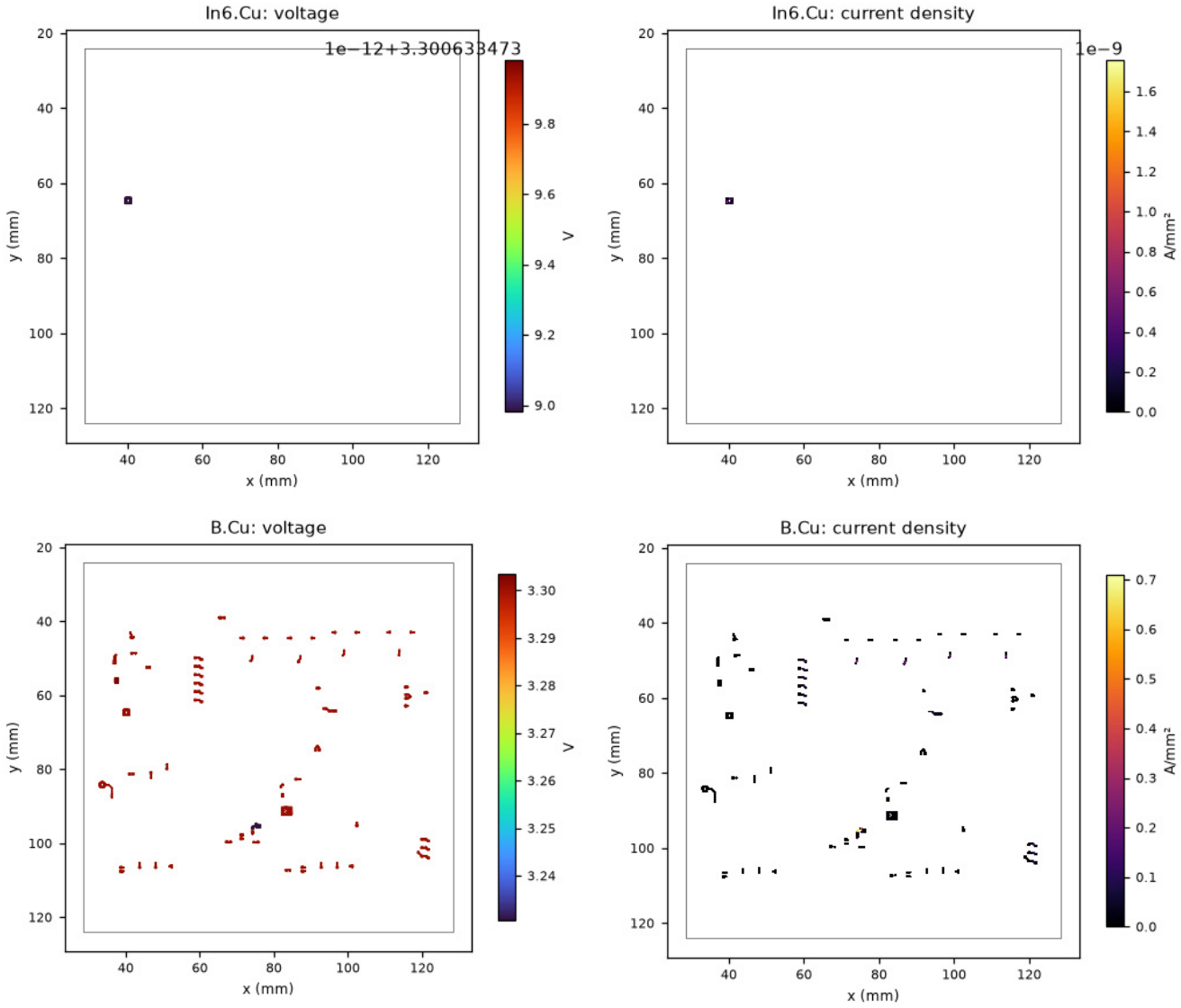

In3.Cu: current density
0.25
0.20
0.15
0.10
0.05
0.00
A/mm²
y (mm)
x (mm)


In4.Cu: voltage
1e−12+3.300633473
9.8
9.6
9.4
9.2
9.0
V
y (mm)
x (mm)


In4.Cu: current density
1e−9
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
A/mm²
y (mm)
x (mm)


In5.Cu: voltage
1e−12+3.300633473
9.8
9.6
9.4
9.2
9.0
V
y (mm)
x (mm)


In5.Cu: current density
1e−9
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
A/mm²
y (mm)
x (mm)



Current-density maps clip at the 99.5th percentile; pad corners are singular and would otherwise wash out the map.

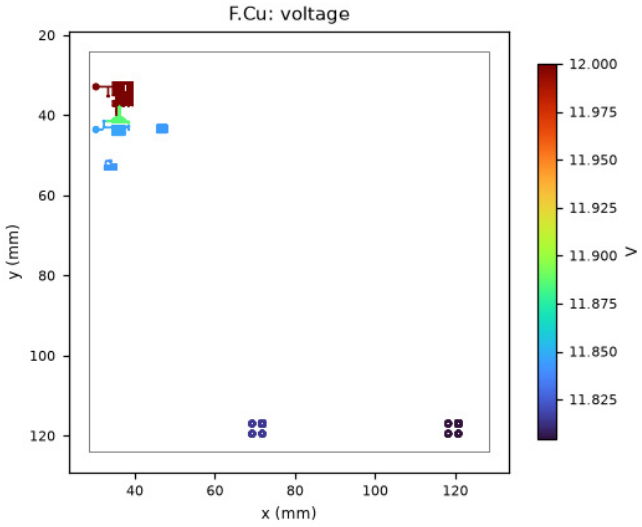
+12V (+2 nets)

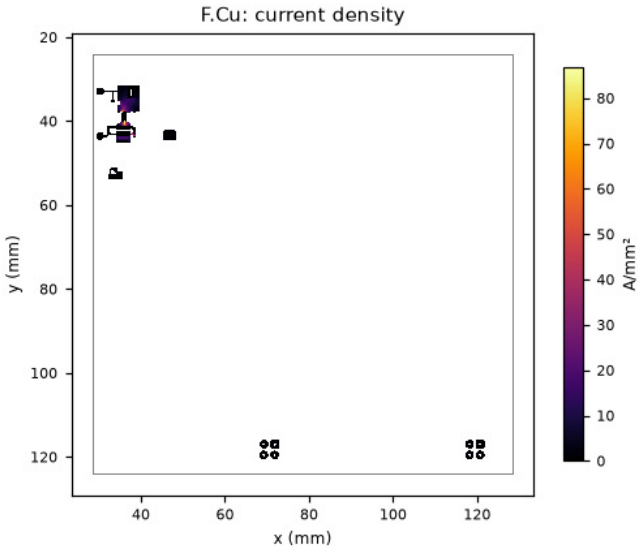
driven by 12V; nets: +12V, +12Vin, Net-(U201-OUT).

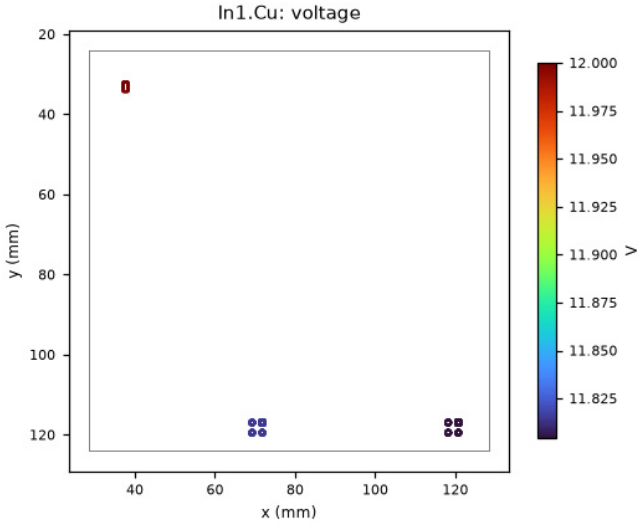
Interactive 3D view: Audio_Visualiser_DSP_kipi_report_2026-10-18_211614_3d.html

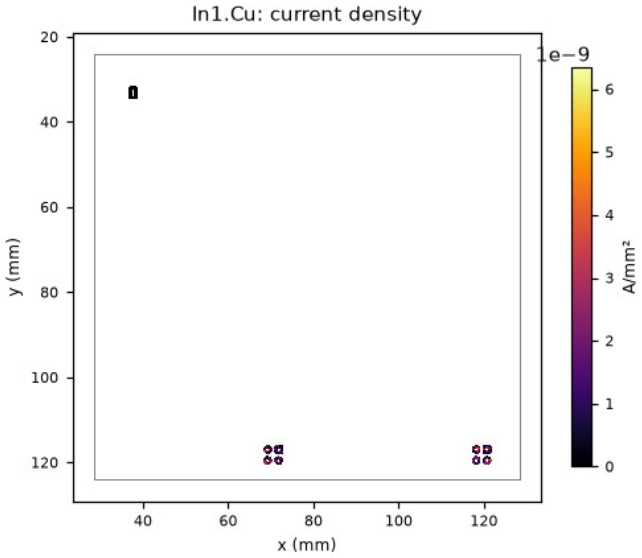
Layer	Copper (mm²)	V min	V max	Max drop (mV)	Max boost (mV)	|J| p99.5 (A/mm²)	|J| max	Loss (mW)
F.Cu	69.3	11.804031	12.000000	195.969	0.000	86.8	229	7.433
In1.Cu	16.4	11.804031	12.000000	195.969	0.000	6.35e-09	9.66e-09	0.000
In2.Cu	53.1	11.804031	12.000000	195.969	0.000	82.9	131	2.326
In3.Cu	2463.5	11.804031	12.000000	195.969	0.000	51.7	105	143.287
In4.Cu	16.4	11.804031	12.000000	195.969	0.000	6.35e-09	9.66e-09	0.000
In5.Cu	16.4	11.804031	12.000000	195.969	0.000	6.35e-09	9.66e-09	0.000
In6.Cu	16.4	11.804031	12.000000	195.969	0.000	6.35e-09	9.66e-09	0.000
B.Cu	101.6	11.804031	12.000000	195.969	0.000	36.1	73.5	2.338

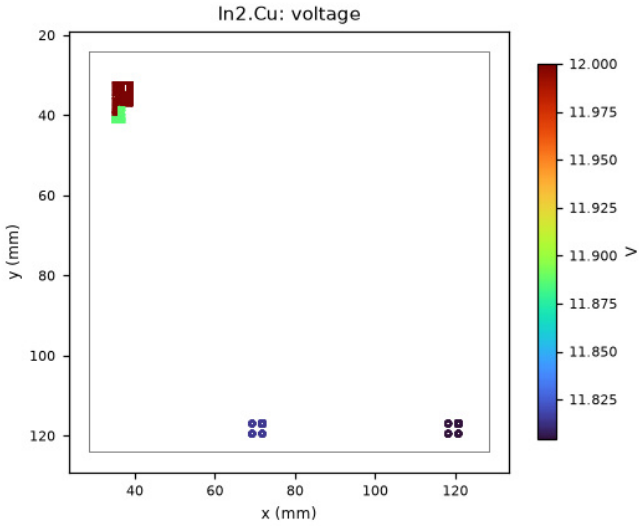
Drop and boost are measured from the 12 V set point.

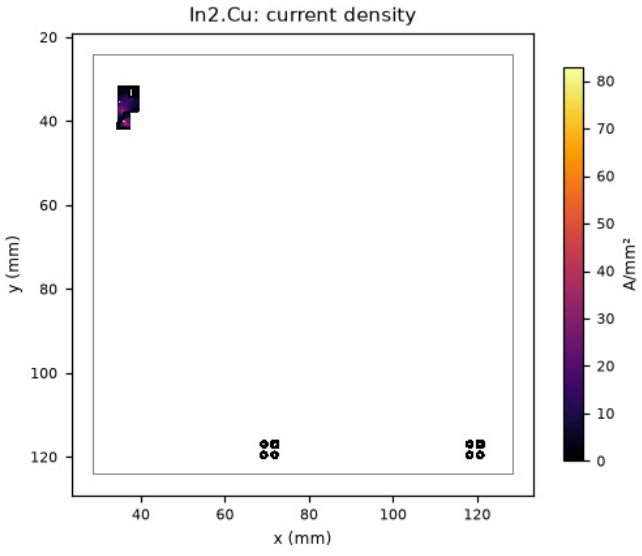

F.Cu: voltage
y (mm)
x (mm)
V
12.000
11.975
11.950
11.925
11.900
11.875
11.850
11.825

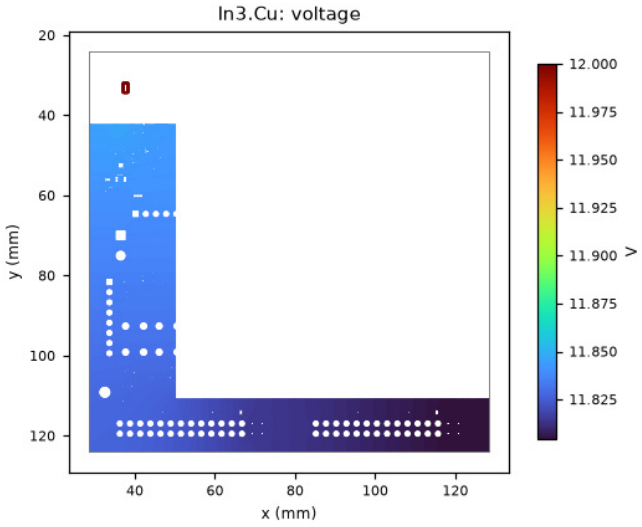

F.Cu: current density
y (mm)
x (mm)
A/mm²

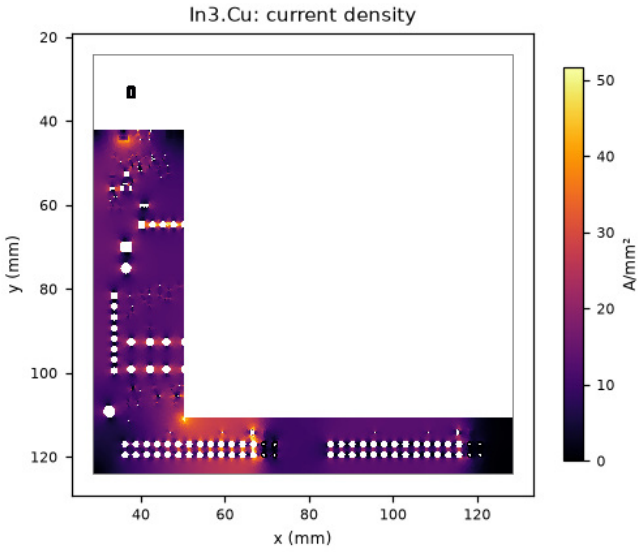

In1.Cu: voltage
y (mm)
x (mm)
V
12.000
11.975
11.950
11.925
11.900
11.875
11.850
11.825

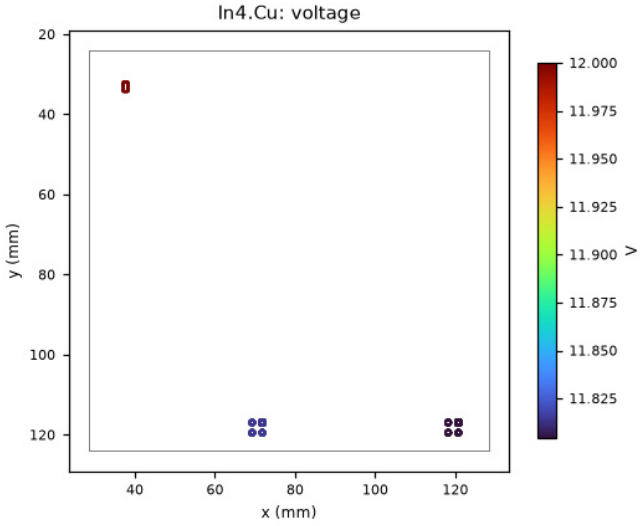

In1.Cu: current density
1e−9
y (mm)
x (mm)
A/mm²

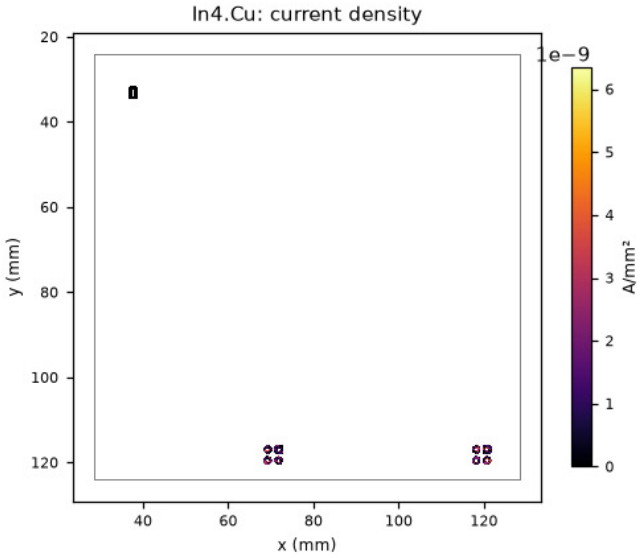

In2.Cu: voltage
y (mm)
x (mm)
V
12.000
11.975
11.950
11.925
11.900
11.875
11.850
11.825

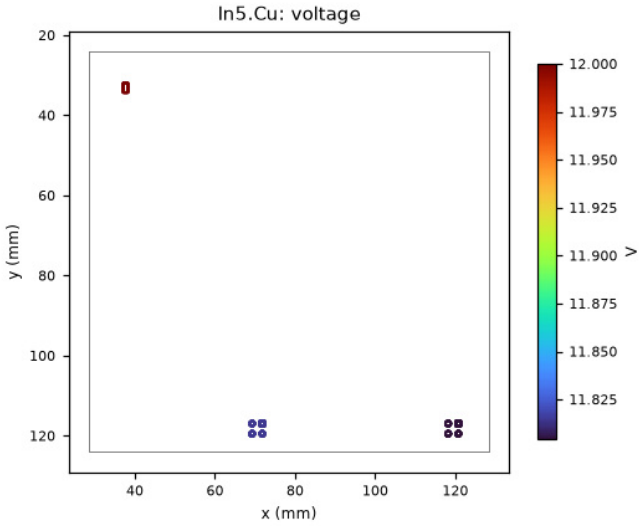

In2.Cu: current density
y (mm)
x (mm)
A/mm²

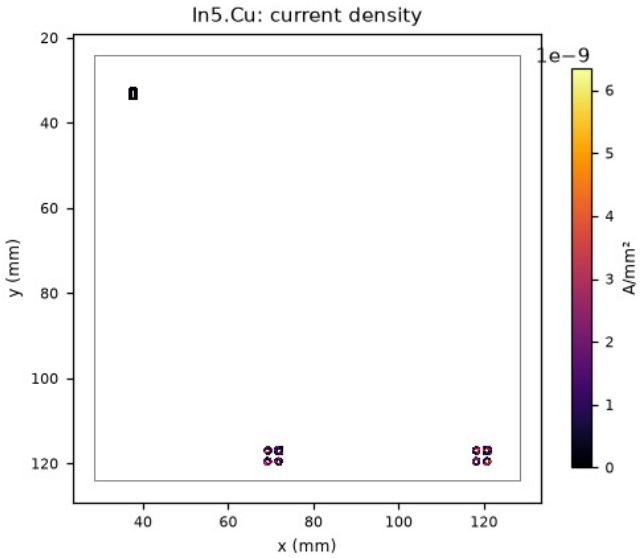

In3.Cu: voltage
y (mm)
x (mm)
V

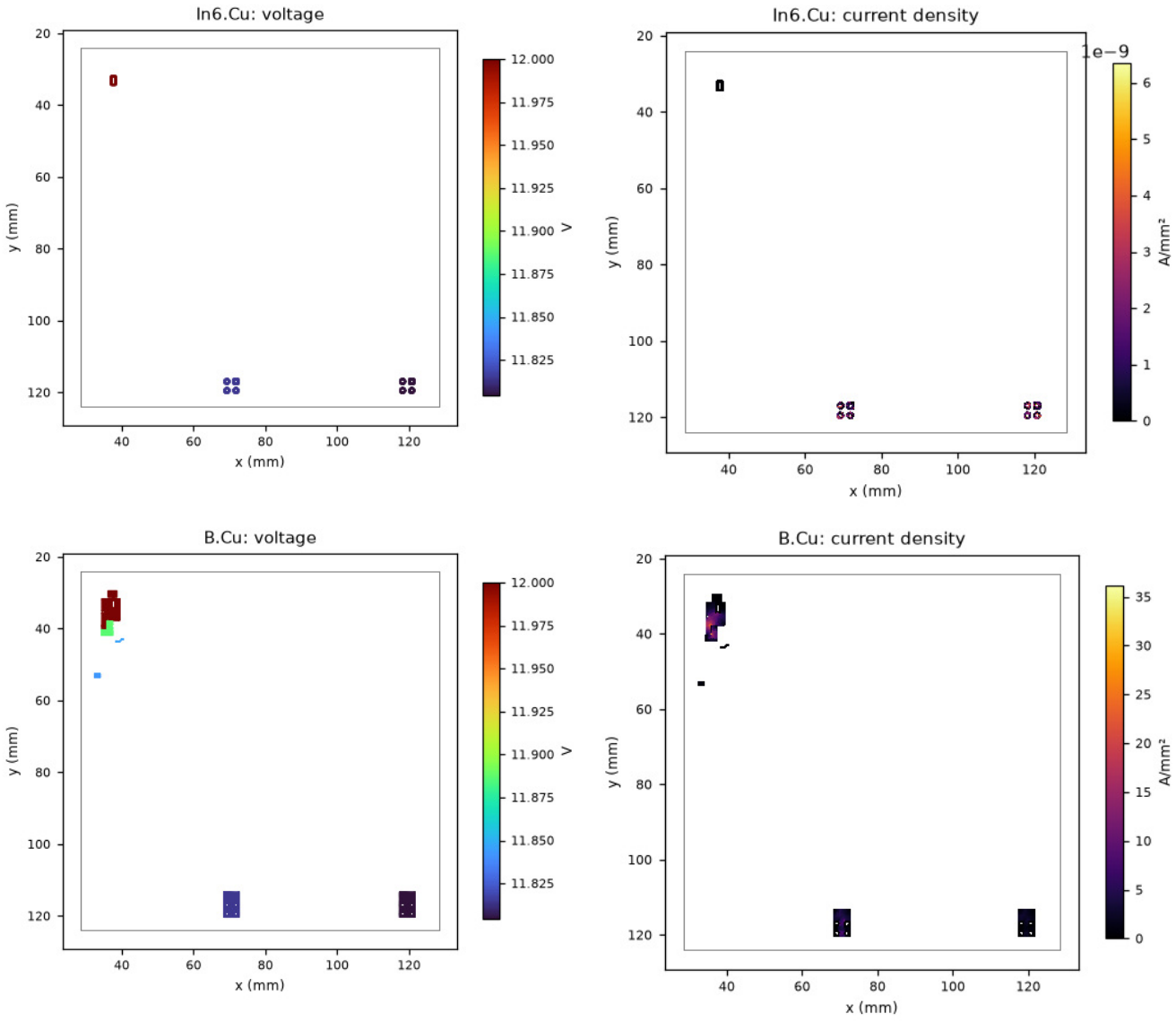

In3.Cu: current density
y (mm)
x (mm)
A/mm²


In4.Cu: voltage
y (mm)
x (mm)
V


In4.Cu: current density
1e−9
y (mm)
x (mm)
A/mm²


In5.Cu: voltage
y (mm)
x (mm)
V


In5.Cu: current density
1e−9
y (mm)
x (mm)
A/mm²



Current-density maps clip at the 99.5th percentile; pad corners are singular and would otherwise wash out the map.

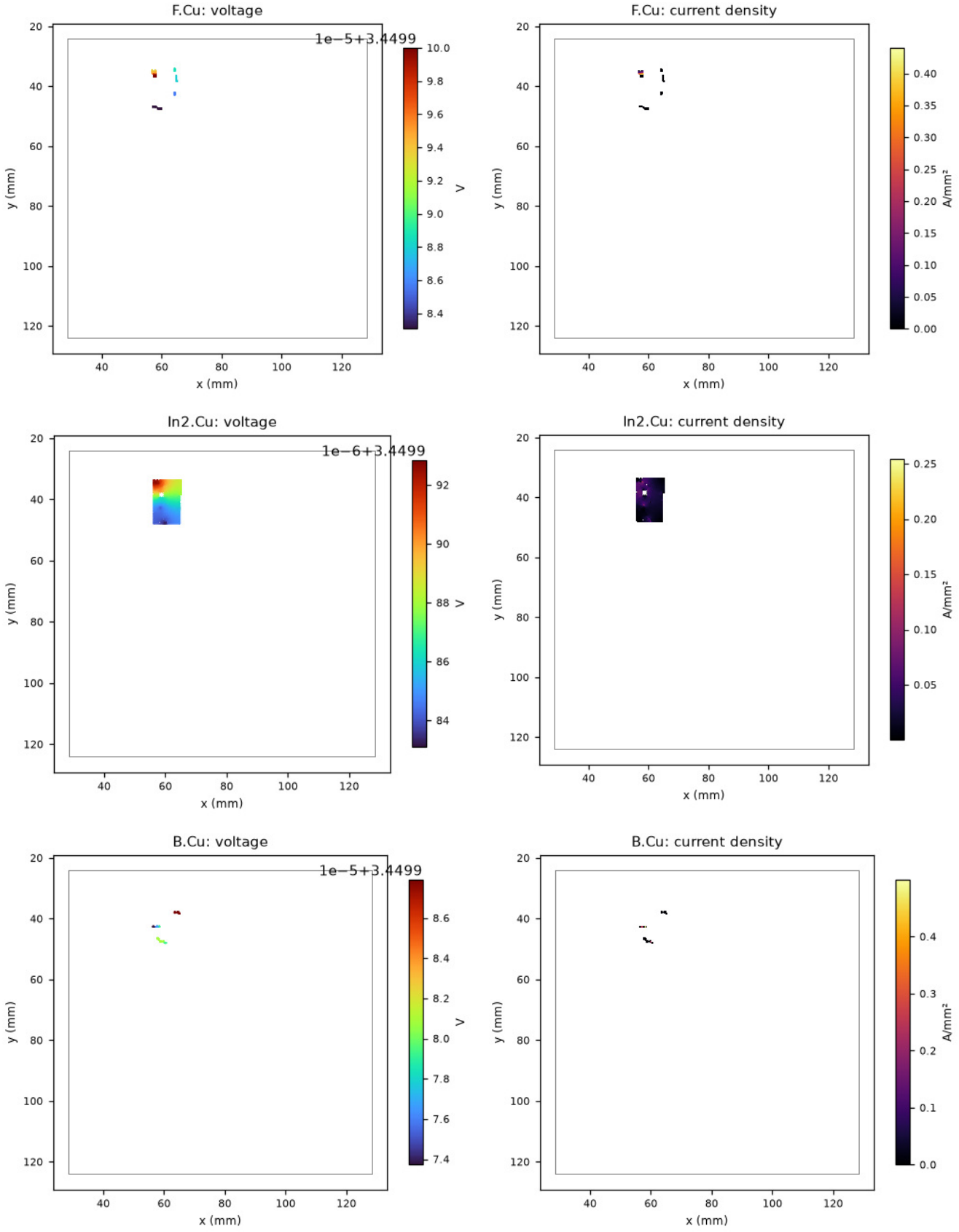
+3.3V_USB

driven by +3.3V_USB; nets: +3.3V_USB.

Interactive 3D view: Audio_Visualiser_DSP_kipi_report_2026-10-18_211614_3d.html

Layer	Copper (mm²)	V min	V max	Max drop (mV)	Max boost (mV)	|J| p99.5 (A/mm²)	|J| max	Loss (mW)
F.Cu	3.6	3.449983	3.450000	0.017	0.000	0.44	0.603	0.000
In2.Cu	118.3	3.449983	3.449993	0.017	0.000	0.254	0.321	0.000
B.Cu	2.6	3.449974	3.449988	0.026	0.000	0.5	0.598	0.000

Drop and boost are measured from the 3.45 V set point.



Current-density maps clip at the 99.5th percentile; pad corners are singular and would otherwise wash out the map.

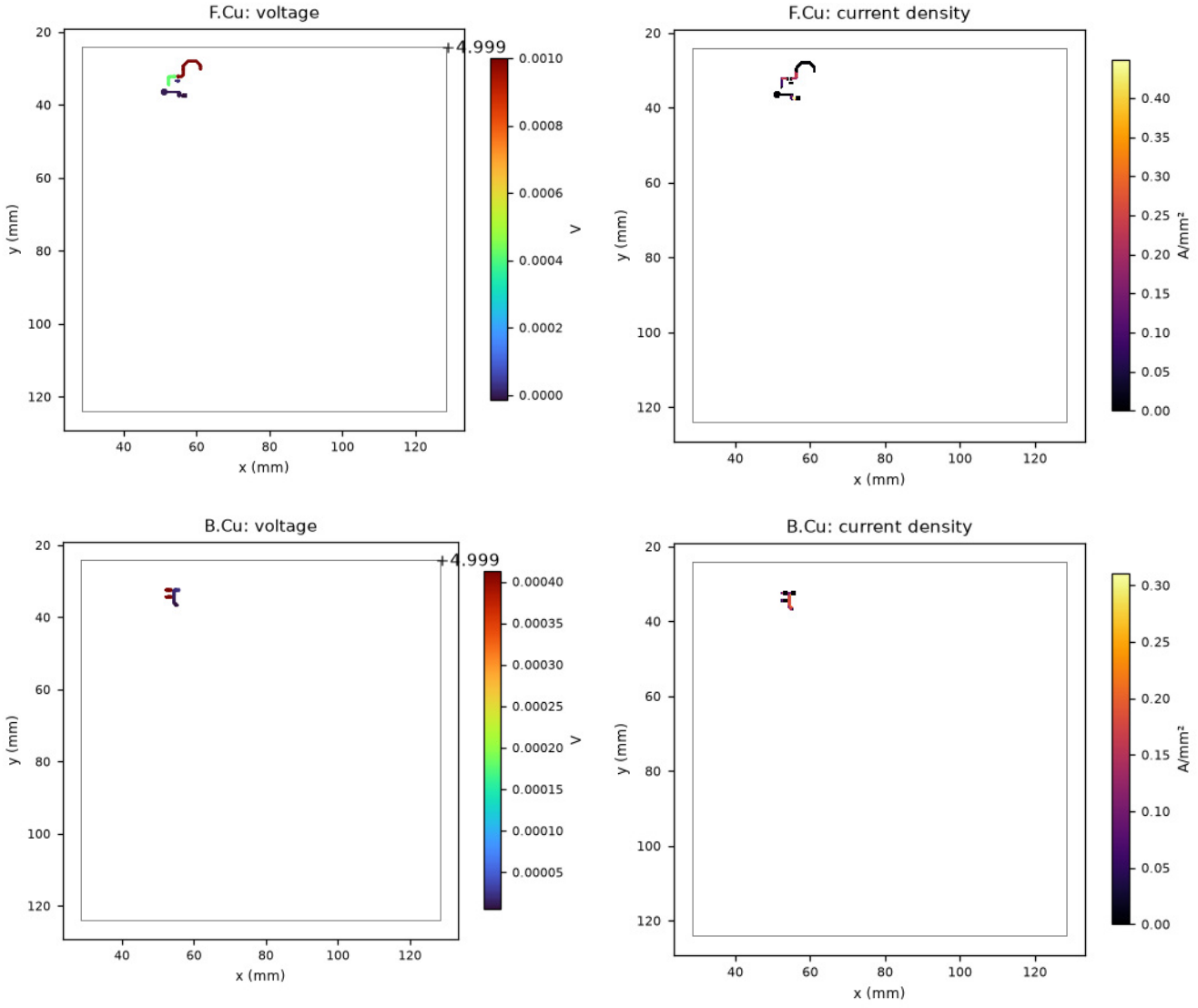
+5V_USB (+2 nets)

driven by +5V_USB; nets: +5V_USB, Net-(FB901-Pad2), Net-(U904-EN).

Interactive 3D view: Audio_Visualiser_DSP_kipi_report_2026-10-18_211614_3d.html

Layer	Copper (mm²)	V min	V max	Max drop (mV)	Max boost (mV)	|J| p99.5 (A/mm²)	|J| max	Loss (mW)
F.Cu	11.6	4.998984	5.000000	1.016	0.000	0.449	0.804	0.000
B.Cu	5.8	4.999005	4.999413	0.995	0.000	0.311	0.557	0.000

Drop and boost are measured from the 5 V set point.



Current-density maps clip at the 99.5th percentile; pad corners are singular and would otherwise wash out the map.

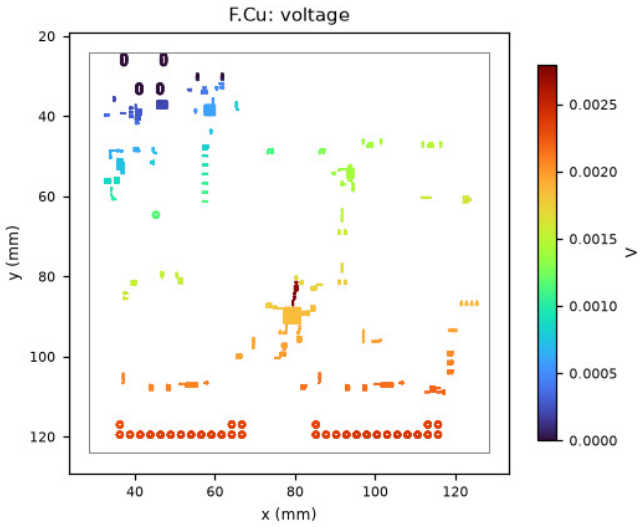
GND (+1 net)

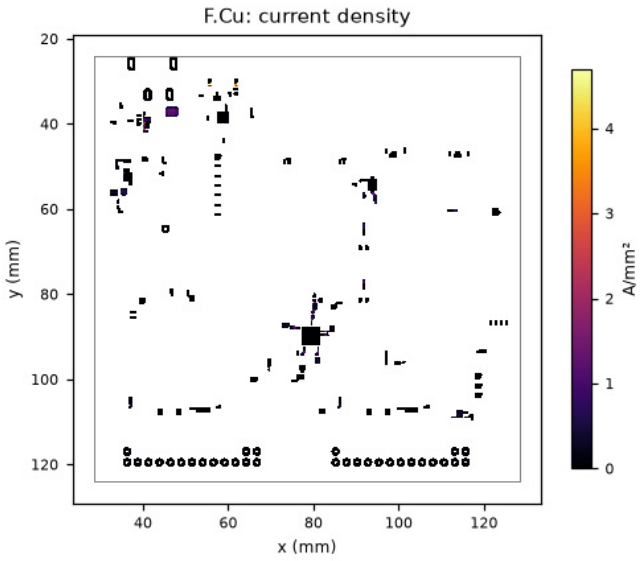
ground, referenced at 12V, +5V_USB; nets: GND, GNDA_DSP.

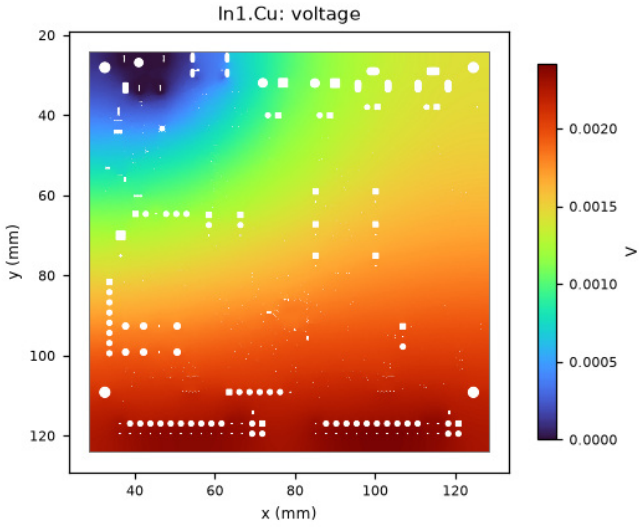
Interactive 3D view: Audio_Visualiser_DSP_kipi_report_2026-10-18_211614_3d.html

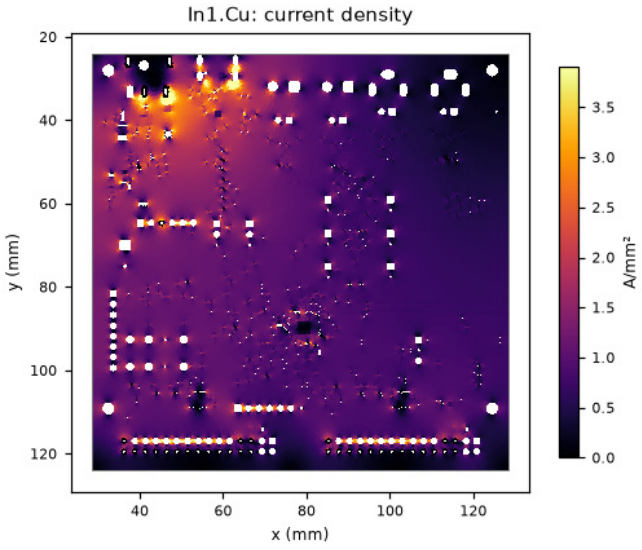
Layer	Copper (mm²)	V min	V max	Max below 0 V (mV)	Max above 0 V (mV)	|J| p99.5 (A/mm²)	|J| max	Loss (mW)
F.Cu	175.3	-0.000000	0.002795	0.000	2.795	4.7	47.5	0.131
In1.Cu	9274.1	0.000000	0.002416	0.000	2.416	3.9	40.1	3.111

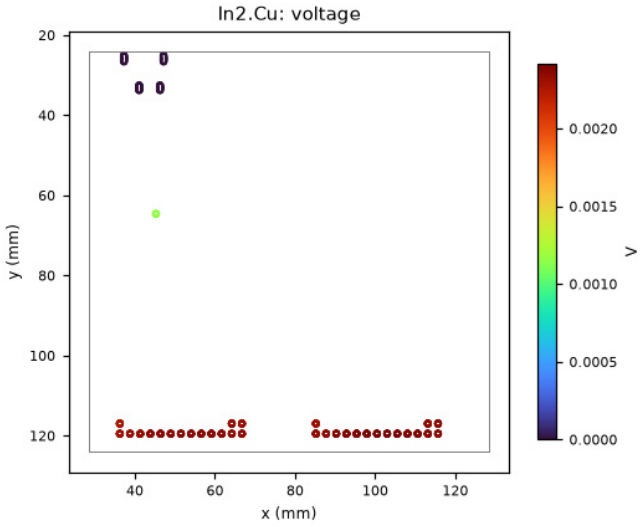
Layer	Copper (mm²)	V min	V max	Max below 0 V (mV)	Max above 0 V (mV)	|J| p99.5 (A/mm²)	|J| max	Loss (mW)
In2.Cu	61.1	0.000000	0.002416	0.000	2.416	1.3e-12	2.17e-12	0.000
In3.Cu	61.1	0.000000	0.002416	0.000	2.416	1.3e-12	2.17e-12	0.000
In4.Cu	9274.1	0.000000	0.002416	0.000	2.416	3.44	16.6	2.982
In5.Cu	61.1	0.000000	0.002416	0.000	2.416	1.3e-12	2.17e-12	0.000
In6.Cu	9274.1	0.000000	0.002416	0.000	2.416	3.45	11.7	2.970
B.Cu	187.3	-0.000000	0.002416	0.000	2.416	1.41	2.5	0.013

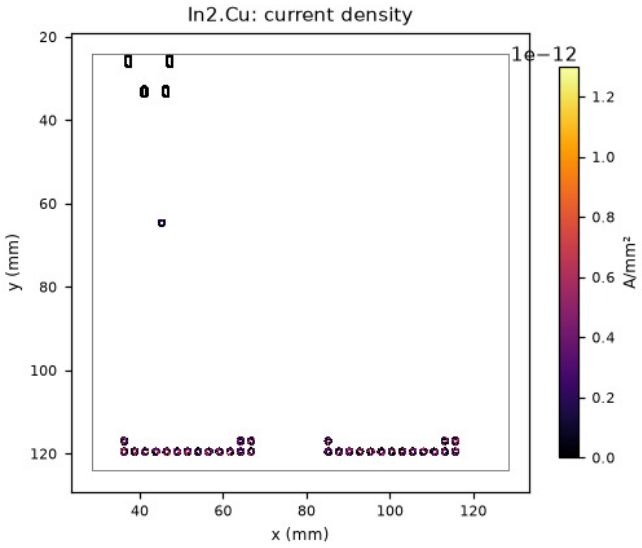


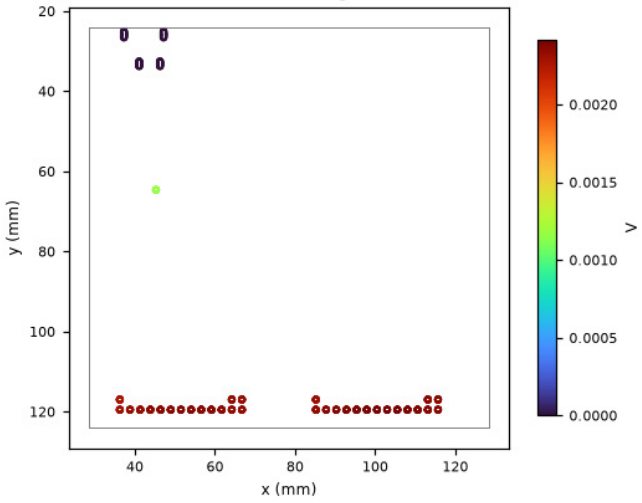


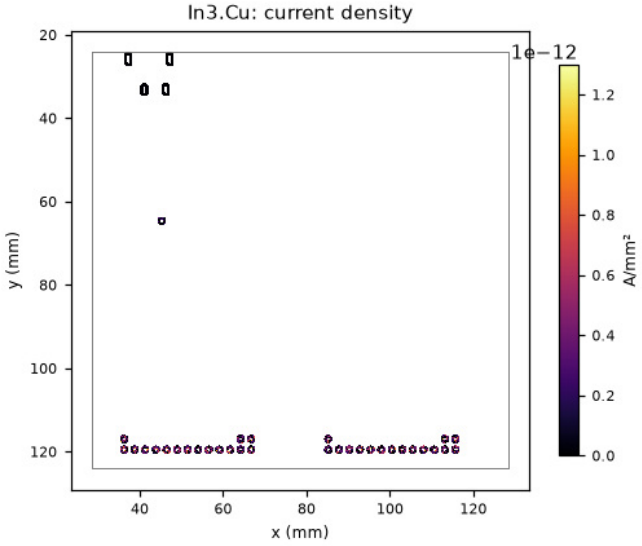


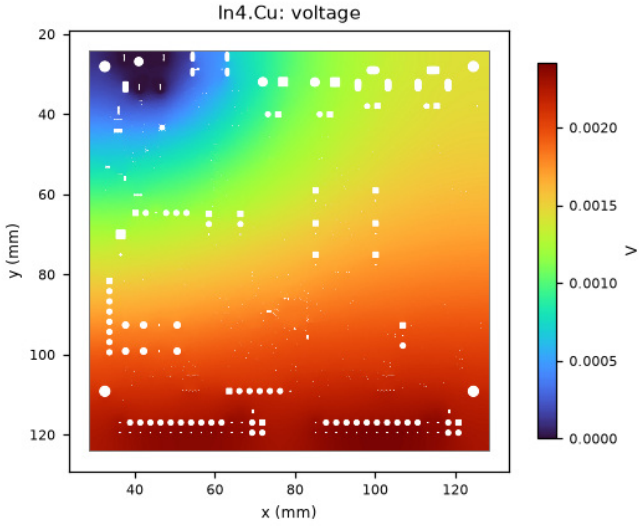


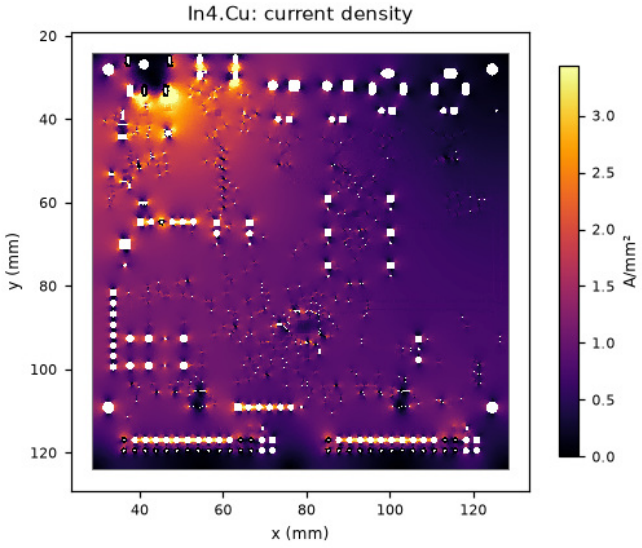

In2.Cu: voltage
y (mm)
x (mm)
V

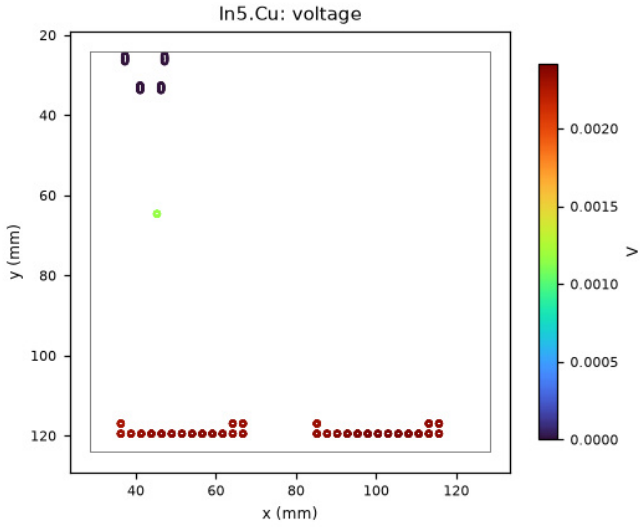

In2.Cu: current density
1e−12
y (mm)
x (mm)
A/mm²

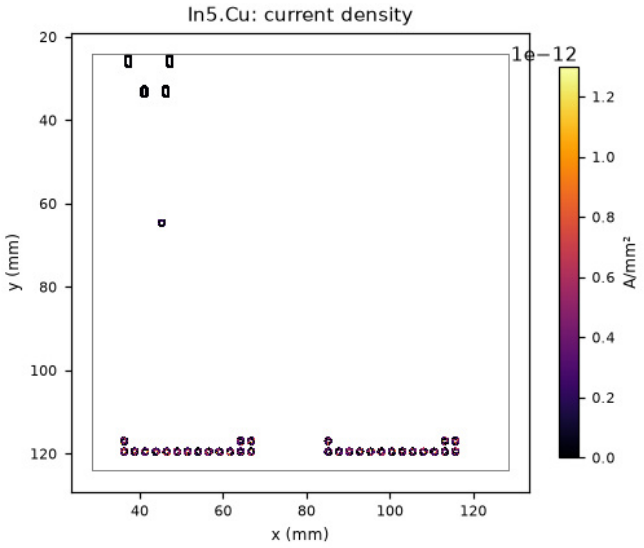

In3.Cu: voltage
y (mm)
x (mm)
V

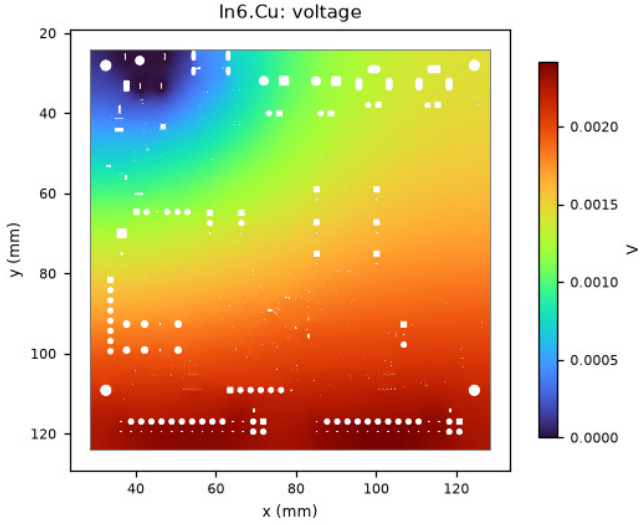

In3.Cu: current density
1e−12
y (mm)
x (mm)
A/mm²

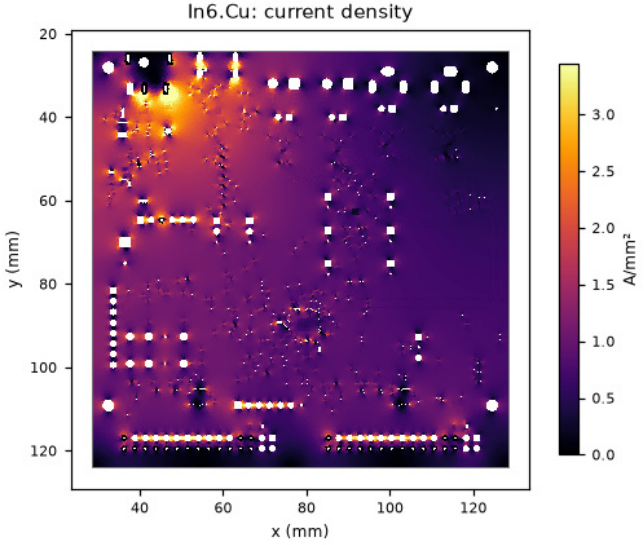

In4.Cu: voltage
y (mm)
x (mm)
V

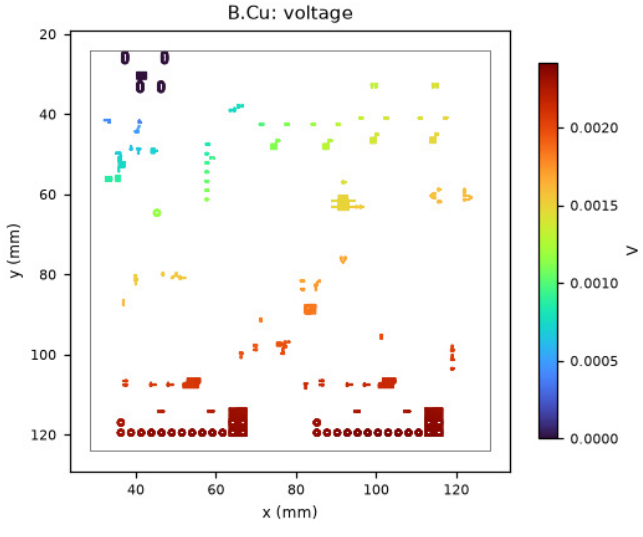

In4.Cu: current density
y (mm)
x (mm)
A/mm²

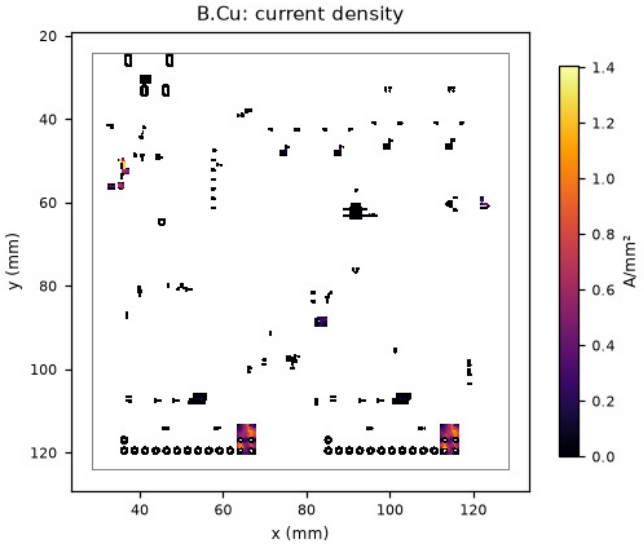












Current-density maps clip at the 99.5th percentile; pad corners are singular and would otherwise wash out the map.

User notes